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Maternal age enhances purifying selection on pathogenic mutations in complex I genes of mammalian mtDNA

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Mitochondrial diseases, caused mainly by pathogenic mitochondrial DNA (mtDNA) mutations, pose major challenges due to the lack of effective treatments. Investigating the patterns of maternal transmission of mitochondrial diseases could pave the way for preventive approaches. In this study, we used DddA-derived cytosine base editors (DdCBEs) to generate two mouse models, each haboring a single pathogenic mutation in complex I genes (*ND1* and *ND5*), replicating those found in human patients. Our findings revealed that both mutations are under strong purifying selection during maternal transmission and occur predominantly during postnatal oocyte maturation, with increased protein synthesis playing a vital role. Interestingly, we discovered that maternal age intensifies the purifying selection, suggesting that older maternal age may offer a protective effect against the transmission of deleterious mtDNA mutations, contradicting the conventional notion that maternal age correlates with increased transmitted mtDNA mutations. As collecting comprehensive clinical data is needed to understand the relationship between maternal age and transmission patterns in humans, our findings may have profound implications for reproductive counseling of mitochondrial diseases, especially those involving complex I gene mutations.

Mitochondria, the powerhouse of the cell and important signaling hubs, harbor their own genome (mtDNA) composed of 37 genes, including two mt-ribosome RNAs, 22 mt-tRNAs and 11 mt-mRNAs (two are bicistronic), encoding 13 proteins that are core subunits of the oxidative phosphorylation (OXPHOS) system[1,2]. Pathogenic mutations in mtDNA can cause mitochondrial diseases, with a prevalence of approximately one in 5,000. Patients with such diseases typically show heteroplasmy of mtDNA, where wild-type and mutant mtDNA coexist within a single cell. Most mutations in mtDNA arise as errors during mtDNA replication[3,4], and, once formed, they may undergo maternal transmission to affect multiple generations[5]. Currently, approximately 100 mtDNA mutations have been confirmed to cause mitochondrial diseases (https://www.mitomap.org/MITOMAP). However, there are substantial variations in the clinical incidence rates of these mutations. Only a small number of specific sites exhibit higher frequencies, and most sites are relatively rare in the clinic.

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The patterns governing the maternal transmission of these pathogenic mutations remain unclear.

The mtDNA is transmitted maternally—that is, passed down through the female germline—in mammals[6], and only a subset of the pool of mtDNA molecules present in the oocyte will be transmitted to the offspring due to a bottleneck effect in the early stages of embryogenesis[7–9]. Evidence from human pedigrees suggests that certain mtDNA variants are subject to positive or negative selection during transmission, which could influence the inheritance of mitochondrial diseases[6,10,11]. However, how the selection acts on human pathogenic mutations and influences the transmission of mitochondrial diseases, as well as the potential rules governing this selection process, remain unknown. This knowledge gap is mainly attributable to the historical lack of effective editing tools for generating mammalian models carrying the same mutant mtDNA homologous to human mutants causing mitochondrial diseases. Recent technological advancements using DddA-derived cytosine base editors (DdCBEs) have enabled targeted editing of specific mtDNA sites[12,13], offering opportunities for creating such mammalian models[14–16]. These models are crucial for investigating the transmission patterns of human mitochondrial diseases as well as for understanding the factors that influence selection during transmission.

Among all the human pathogenic mtDNA mutations, those affecting complex I of the respiratory chain play an important role. Complex I, the largest of the four respiratory chain complexes, is the main point where electrons enter the respiratory chain[17,18]. mtDNA mutations in complex I genes are associated with a wide range of human diseases, including neurodegenerative disorders; cardiovascular, skeletal muscle and liver diseases; and more. These mutations make up the largest proportion of all confirmed pathogenic mtDNA mutations in protein-coding genes (from MITOMAP; 38/63 = 60.31%). Currently, no direct treatment exists for mitochondrial diseases caused by complex I deficiency. This highlights the importance of studying the inheritance pattern to find a strategy to prevent transmission of these diseases.

Several previous studies on pathogenic mtDNA mutations in mice and *Drosophila* yielded valuable insights into this field. In mice, selection prevents the birth of pups with high levels of the mtDNA mutation in the *tRNA-Met* and *Ala* genes[19–22], and selection rapidly removes a heteroplasmic mtDNA mutation in the gene encoding ND6 after transmission for only a few generations[23]. In *Drosophila*[24–27], selection eliminates *Cox1* mutation (a temperature-sensitive lethal mutation in the complex IV gene) during early oogenesis when exposed to a restrictive temperature of 29 °C. Two potential selection mechanisms were proposed from these models: germline mitophagy[27] and the dominance of healthy mitochondria under replication stress[24,28–30]. However, it remains uncertain whether these mechanisms are universally applicable to various types of mutations in mammals, particularly those responsible for human mitochondrial diseases.

To investigate the transmission patterns of mitochondrial disease caused by mtDNA mutations in complex I genes, we employed DdCBE to generate two mouse models. Each model carried a single point mutation in complex I genes (*ND1* and *ND5*), homologous to mutants causing human mitochondrial diseases. Our findings revealed that purifying selection took place in both models to prevent the pathogenic mutants into the offspring. The selection occurred mainly during postnatal oocyte maturation, initiating after the primordial follicle stage and ending before the first meiosis was completed. It predominantly acted at the organelle level, with increased translation activity playing a vital role. Notably, such purifying selection was enhanced as maternal age increases, implying that maternal age may serve as a protective factor against transmitting these mtDNA mutations. This finding challenges the widely held notion that increased maternal age leads to negative effect with a higher transmission of mtDNA mutations to the next generation[11,31,32] and warrants further clinical attention to the maternal age effects for transmitted mitochondrial diseases. This knowledge may have important implications for clinical reproductive counseling, particularly for mitochondrial diseases caused by mtDNA mutations of complex I genes.

Results

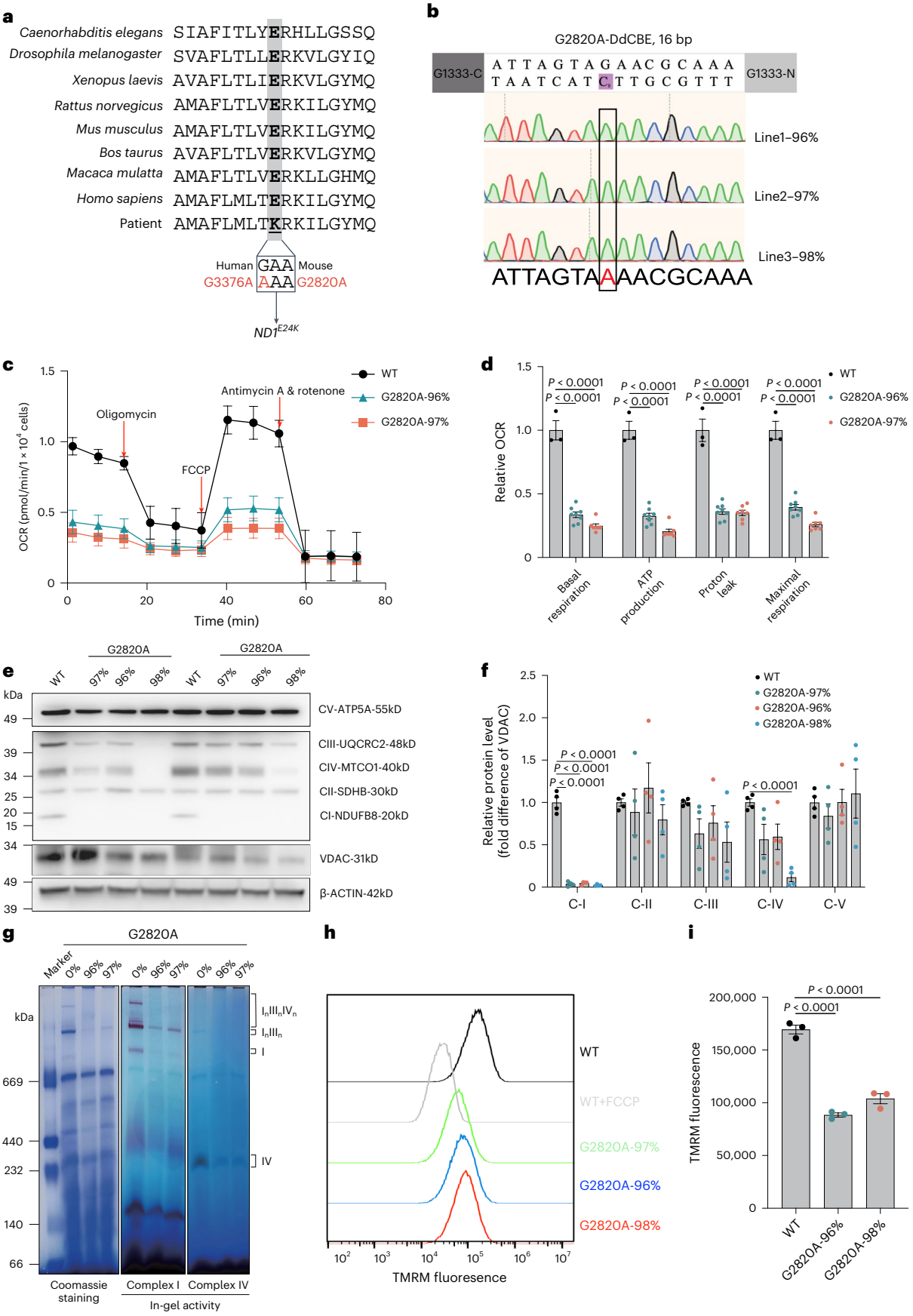
ND1 point mutation disrupts mitochondrial complex I in mouse embryonic fibroblasts

Because there are no available mtDNA mutant mouse models with homologous mutation to human mitochondrial diseases in complex I genes to investigate maternal transmission, we used DdCBE, the newly developed tool for mtDNA editing, to construct appropriate mouse models. We chose the 2820 site of the mouse mtDNA for editing, which is homologous to the human 3376 site (Fig. 1a), a point mutation that can cause human mitochondrial disease[33]. Specifically, this mutation results in the substitution of the 24th amino acid of the human and mouse ND1 protein, changing it from glutamic acid (E) to lysine (K). We initially used mouse embryonic fibroblasts (MEFs) to set up the mouse editing system and assess whether the pathogenesis observed in the model could mimic that of the homologous human mutations. To accurately assess the obtained editing efficiency, we modified the two plasmids with either eGFP or mCherry to isolate mutant cells based on dual-fluorescence positivity by fluorescence-activated cell sorting (FACS) (Extended Data Fig. 1a,b). Their mutation load, as determined by sequencing, represents the editing efficiency. After FACS of MEFs transfected with modified editing plasmids, we obtained six different single-cell-derived MEF lines (hereafter, *ND1* mutant MEFs). Strikingly, the obtained cell lines all showed a very high mutation load for the 2820 site according to Sanger sequencing (Fig. 1b). To precisely determine the mutation load of the edited cells, we employed the highly accurate Hi-TOM sequencing method[34], which has been validated with high accuracy (Methods and Extended Data Fig. 1c). With this method, we found that the mutation loads ranged from 93% to 98% in the six cell lines derived from FACS-sorted single cells (Extended Data Fig. 1d). Thus, our editing system works efficiently on the 2820 site in MEFs.

To assess whether the *ND1* mutant MEFs exhibit mitochondrial defects, we tested their mitochondrial respiration activity using a Seahorse XFe96 Analyzer. We found a significant decrease in the oxygen consumption rate (OCR) in the mouse cells with high *ND1* mutation loads (above 96%) (Fig. 1c,d). Furthermore, when we studied the effect

Fig. 1 | The mouse mtDNA 2820 G > A mutation causes mitochondrial dysfunction. a, Alignment of ND1 protein sequences (amino acids 16–32) across various species, highlighting the conserved glutamine (E) at position 24 in bold with a gray background. Mouse G2820A and human G3376A mutations result in an E to K (lysine) substitution in the ND1 protein. **b**, Target region in the mouse *ND1* gene for optimized editing plasmids, with representative editing efficiency shown for three single-cell-derived MEF lines. **c**, OCR in MEF lines with *ND1* mutations at 96% and 97% compared to wild-type (WT) lines, using Seahorse bioenergetic analysis. Sample sizes: WT $n = 3$, G2820A-96% $n = 8$, G2820A-97% $n = 7$. Data are mean ± s.e.m., replicated independently ≥3 times. **d**, Analysis of respiratory parameters from **c**. Data are presented as mean ± s.e.m. *P* values were determined by Kruskal–Wallis test and Dunn's post hoc analysis. **e**, Representative western blot image of OXPHOS protein levels in control and *ND1* mutant MEF cells. The experiment was repeated four times. **f**, Quantitative analysis of OXPHOS protein levels from **e**. The intensity of the corresponding band was quantified with ImageJ and normalized to VDAC. Then, all the data were normalized to the WT groups. Complexes are denoted as C-I to C-V. For each, $n = 4$, presented as mean ± s.e.m. *P* values were determined by one-way ANOVA with Tukey's comparisons. **g**, In-gel enzyme activities of complexes I and IV in WT and mutant MEF lines after BN-PAGE (400 μg per lane). Coomassie staining was used as the loading control. The experiment was repeated three times. **h**,**i**, MMP in WT and *ND1* mutant MEFs assessed by TMRM fluorescence intensity using FACS in **h** and quantitatively analyzed in **i**, $n = 3$ per group. Data are mean ± s.e.m. *P* values were determined by one-way ANOVA followed by Tukey's test.

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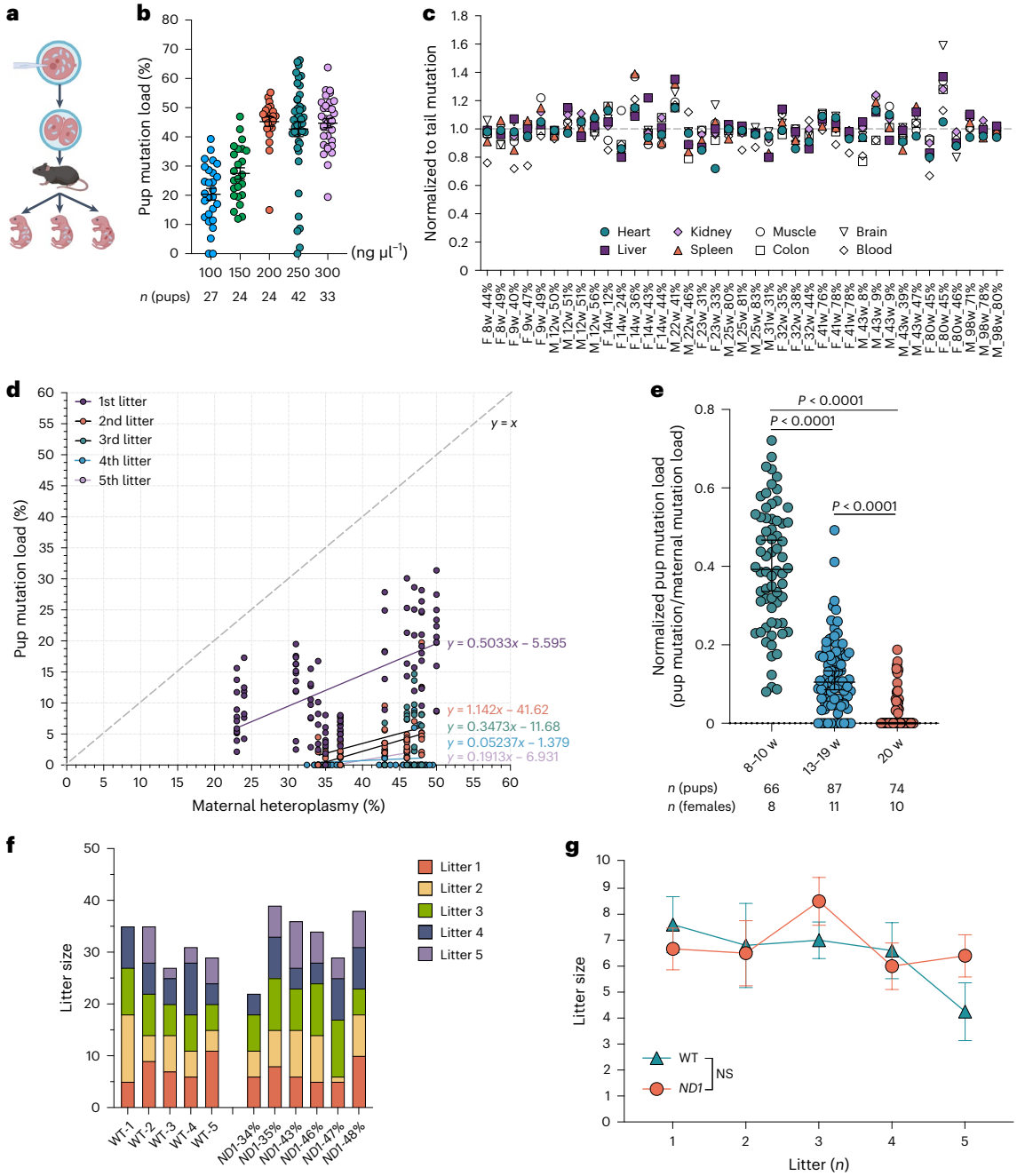


Fig. 2 | Age enhances the purifying selection of m.2820 G > A mtDNA during transmission. a, Schematic diagram of the construction of *ND1* 2820 G > A mutant mice. Created with BioRender. **b**, In vivo editing efficiency represented by the mutation load of newborn pups. The *x* axis indicates the concentration of injected editing mRNA; the *y* axis indicates the mutation load of newborn pups. Each open circle on the graph represents an individual animal. The total number of pups born from different mRNA concentrations is listed along the *x* axis. Data are mean ± s.e.m. **c**, Normalized mutation loads in organs and blood from *ND1* mutants across ages (8–9 weeks, $n = 5$; 12–14 weeks, $n = 9$; 22–25 weeks, $n = 7$; 31–32 weeks, $n = 4$; 41–43 weeks, $n = 8$; 80 weeks, $n = 3$; 98 weeks, $n = 3$). 'F' indicates females; 'M' indicates males. Normalization is relative to tail mutation load at euthanization. **d**, Correlation of pup mutation loads ($n = 274$) with maternal mutation loads ($n = 12$), by litter. Different litters are color-coded. The y = x line shows expected transmission without selection, and y =ax + b is the linear regression of the mutation loads of pups from different females but produced in the same litter order. **e**, Relationship between maternal age at conception and pup normalized mutation load. Normalization is pup toe mutation load divided by maternal toe mutation load. Data are median ± 95% CI. *P* values were determined by Kruskal–Wallis test and Dunn's comparison. **f**,**g**, Comparison of litter sizes between WT and *ND1* mutant females (**f**) and detailed analysis considering litter order (**g**). WT: $n = 5$; *ND1* mutant: $n = 6$. Mean ± s.e.m. is shown, with two-tailed unpaired *t*-test for *P* values. WT, wild-type; NS, not significant.

of these mutations with western blot, blue native PAGE (BN-PAGE) and complex I in-gel activity analysis, we found that complex I assembly was disrupted in the mutant cells (Fig. 1e–g). Complex I pumps protons and is a major contributor to the mitochondrial membrane potential (MMP)[35,36]. Consistently, we found that the MMP, as determined by the tetramethylrhodamine methyl ester (TMRM) labeling assay, was significantly decreased in the mutant MEFs (Fig. 1h,i). In summary, by using the DdCBE approach, we obtained MEFs with high levels of the 2820 G > A mutation of mtDNA that causes mitochondrial complex I defect, allowing us to proceed with generating a mutant mouse model.

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ND1 mutant mtDNA undergoes strong purifying selection during transmission

To construct the G2820A mutant mouse model, we applied the DdCBE method to zygotes and transplanted the resulting embryos into pseudo-pregnant wild-type females (Fig. 2a). We used in vitro transcription from the pair of plasmids showing the highest editing efficiency in MEFs to obtain mRNA for injection into the cytosol of fertilized eggs. We injected varying concentrations of mRNA to determine the optimal level inducing efficient editing. The editing efficiency was determined by mtDNA mutation load of the obtained pups. We found that the mutation load of mutant pups increased with the concentration of injected mRNA, reaching a peak at approximately 200–250 ng μl^{-1}. No further significant change was observed in the pup mutation load after mRNA injection with a concentration of 300 ng μl^{-1}. Therefore, we adopted 250 ng μl^{-1} as the optimal injection concentration. The maximum mutation load in these pups was 66.3%, and the average mutation load was as high as approximately 43% (Fig. 2b). Furthermore, the complete sequencing of the entire mtDNA sequence in the edited pups revealed no off-target effects at other mtDNA sites that exceeded 1% (Supplementary Table 1). This finding provides strong evidence supporting the high specificity of this editing approach.

Given that patients in clinical settings often exhibit varying mutation loads across different tissues, with a notably lower load in blood due to negative selection[37,38], we explored whether such variation also occurs at the organ or blood level in our mutant pups. We examined the mutation load in multiple organs (including heart, liver, kidney, spleen, muscle, colon, brain and tail) as well as blood, in both male and female mice aged between 8 weeks and 100 weeks. Through our analysis, we determined that there was no obvious heteroplasmic shifting among the tested samples (Fig. 2c). Furthermore, the mutation load of the same organ remained consistent without noticeable alterations during aging (Extended Data Fig. 2a–h). Notably, we found no evidence of selective pressure against the *ND1* mutation in the blood of these mutant mice (Extended Data Fig. 2h). However, when we analyzed specific immune cell types that were sorted by flow cytometry, an interesting pattern emerged: T cells displayed a slightly lower mutation rate compared to B cells, neutrophils and monocytes (Extended Data Fig. 2i,j). This subtle variation in mutation rates among different immune cell types suggests the presence of a potential negative selective mechanism within the T cell lineage. Similar patterns were previously observed in patients with the A3243G mutation, further supporting our findings[38].

To maintain the mutant line, we performed extensive crosses between *ND1* mutant females with different mutation loads and wild-type B6 males. We found that strong purifying selection during transmission appeared in the first litters of these females. Surprisingly, the strength of this selection continued to increase in subsequent litters, and many of the pups from the 5th litter harbored almost no *ND1* mutations (Fig. 2d). To better understand the correlation between purifying selection and maternal age, we summarized all the information on the relative mutation loads of pups (mutation loads relative to those of their mothers) and the ages of their mothers when they became pregnant (pup birth date - maternal birth date - 3 weeks of pregnancy). Surprisingly, we found that *ND1* mutants gave birth to pups with a very low *ND1* mutation load or even no mutations when the mothers had reached 20 weeks of age (Fig. 2e).

To determine whether the observed strong selection was linked to the loss of mutant pups during pregnancy, we compared the total and individual litter size at the same litter orders of the mutant females to those of B6 wild-type females. Our analysis indicated no significant difference between the two groups in terms of both total and individual litter size (Fig. 2f,g). In summary, we successfully obtained a mouse model with the $ND1^{E24K}$ (G2820A) point mutation using the DdCBE and identified a strong purifying selection against the *ND1* mutation during transmission that further increased with increasing maternal age.

Age-enhanced purifying selection on *ND1* mutation occurs mainly at female germline

Because a decrease in pup mtDNA mutation load with increasing maternal age was unexpected, we further investigated whether this selective process was already concluded in mature oocytes and repeatable in different generations. Thus, we analyzed the mutation load of mature oocytes superovulated from *ND1* mutants with various mutation loads and found that the mutation load in mature oocytes (M2 oocytes) showed a similar trend to that in the offspring (Fig. 3a). The average relative mutation load in M2 oocytes decreased to 65.5% of the mutation load determined in corresponding tail samples collected from mice at 3–4 weeks of age, to 43.11% at approximately 8 weeks of age, to 18.45% at approximately 14 weeks of age and to less than 10% in mice older than 23 weeks. Many M2 oocytes without any *ND1* mutations were found in mice older than 23 weeks (Fig. 3a). In contrast to the mutation load found in the pups of mutant females, several M2 oocytes with high *ND1* mutation loads were found in females older than 23 weeks (Fig. 3a). We speculate that these oocytes probably either could not be fertilized or could not support normal embryogenesis because we did not obtain any pups with such a high mutation load from these aged females. The mutation load from the superovulated M2 oocytes indicates the purifying selection is concluded once mouse oocytes get matured. Furthermore, such maternal age-enhanced selection was observed in not only F0 but also F1 females, whose mutation load was significantly lower than that of F0 females (Extended Data Fig. 3a,b). The mutation load of pups from aged F1 females significantly decreased, with almost no changes in litter size (Extended Data Fig. 3c). The identified purifying selection phenomenon was also observable in the M2 oocytes of F1 females (Extended Data Fig. 3d).

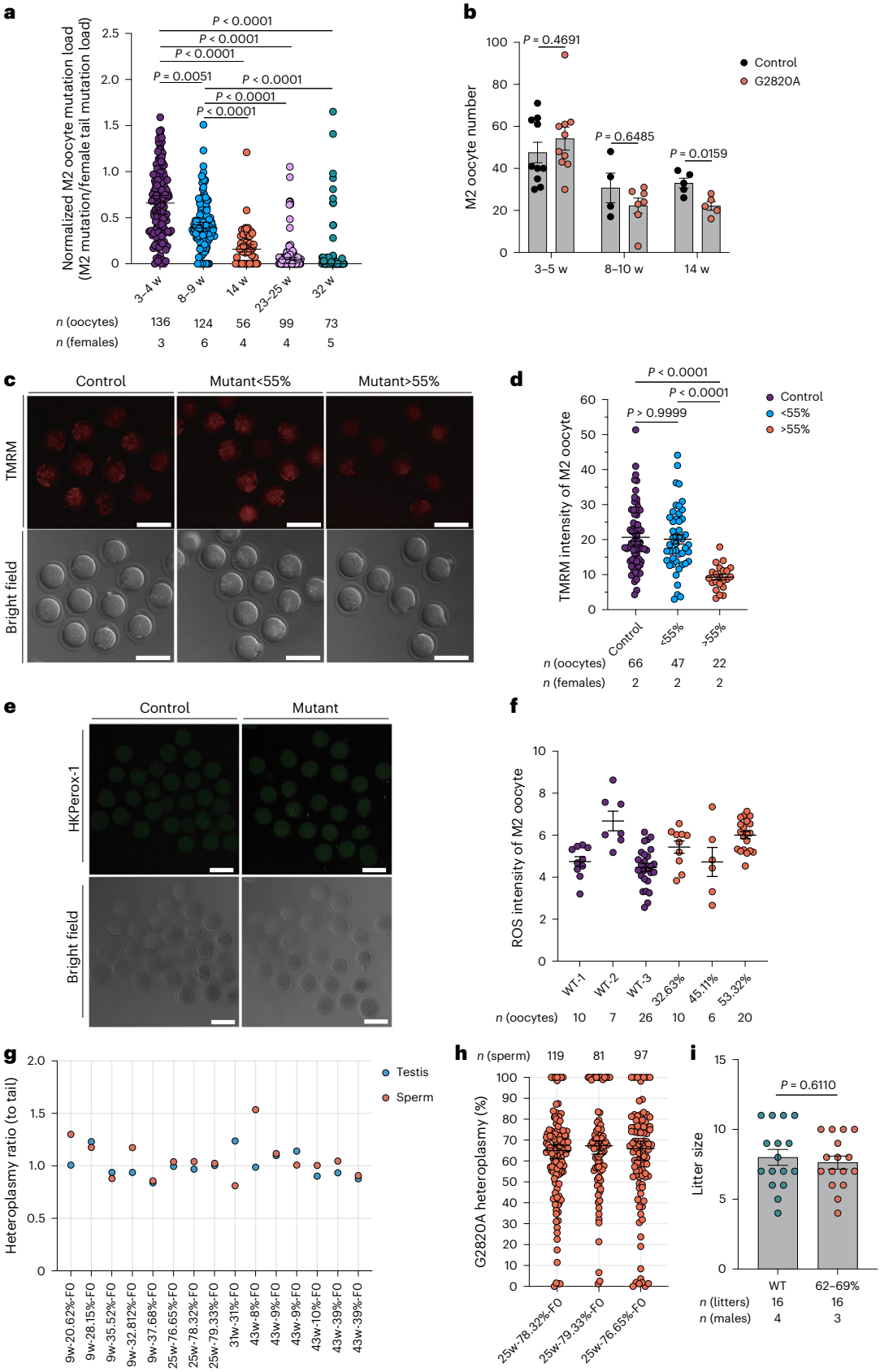
The *ND1* mutation causes severe impairment of mitochondrial respiratory chain function in MEFs; we, therefore, proceeded to investigate whether the mutation is also pathogenic for oocytes because the slightly decreased number of oocytes was observed in superovulated *ND1* mutants at 14 weeks of age (Fig. 3b). To evaluate the effect of *ND1* mutation on mitochondrial function in M2 oocytes, we measured MMP and total hydrogen peroxide (H_2O_2) levels, a main reactive oxygen species (ROS), in oocytes of mutant females. Considering the marked decrease in the mutation load observed in oocytes from adult versus younger females, we evaluated M2 oocytes from 3–4-week-old mice in these analyses. The MMP of M2 oocytes was

Fig. 3 | M2 oocytes exhibit age-enhanced purifying selection against the m.2820 G > A mutation. a, Relationship between maternal age and mutation load in M2 oocytes. M2 oocytes were isolated from each of 22 mutant females at different ages. Each symbol represents one individual oocyte. Data are median ± 95% CI. *P* values were determined by Kruskal–Wallis test with Dunn's comparison. **b**, The numbers of M2 oocytes retrieved from WT control and m.2820 G > A females at different ages. Data are mean ± s.e.m., with group sizes: 3–5w_control ($n = 10$), 3–5w_G2820A ($n = 10$), 8–10w_control ($n = 4$), 8–10w_G2820A ($n = 7$), 14w_control ($n = 5$) and 14w_G2820A ($n = 5$). *P* values were determined by two-tailed Mann–Whitney test. **c**, Representative images of the MMP analysis of M2 oocytes using TMRM staining. Scale bar, 100 µm. **d**, Quantitative analysis of the fluorescence intensity of TMRM in each individual oocyte in **d**. Data are mean ± s.e.m. *P* values were determined by Kruskal–Wallis test with Dunn's comparison. **e**, Representative image of ROS levels in M2 oocytes revealed by HKPerox-1 staining. Scale bar, 100 µm. **f**, Quantitative analysis of the fluorescence intensity in each individual oocyte in **e**. Data are mean ± s.e.m. **g**, Normalized mutation loads of mutant male testes and sperm (normalized to tail mutation load at euthanization). **h**, Mutation loads of individual sperm from mutant males. Data are median ± 95% CI. **i**, Litter size from WT and *ND1* mutant males. Data are mean ± s.e.m. *P* values were determined by two-tailed unpaired *t*-test. w, weeks; WT, wild-type.

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significantly lower in high-mutation-load (>55%) than in wild-type mice or the low-mutation-load (<55%) group (Fig. 3c,d). However, the ROS signal showed no apparent difference between wild-type and mutant oocytes (Fig. 3e,f). This functional analysis shows that the *ND1* mutation impairs mitochondrial function in oocytes with a high mutant level.

Because we observed strong selection against the pathogenic *ND1* mutation during female germline development, we conducted further assessments to determine whether such selection also occurred during male spermatogenesis. The mutation loads found in the testes and sperm of *ND1* mutants were similar to those in the



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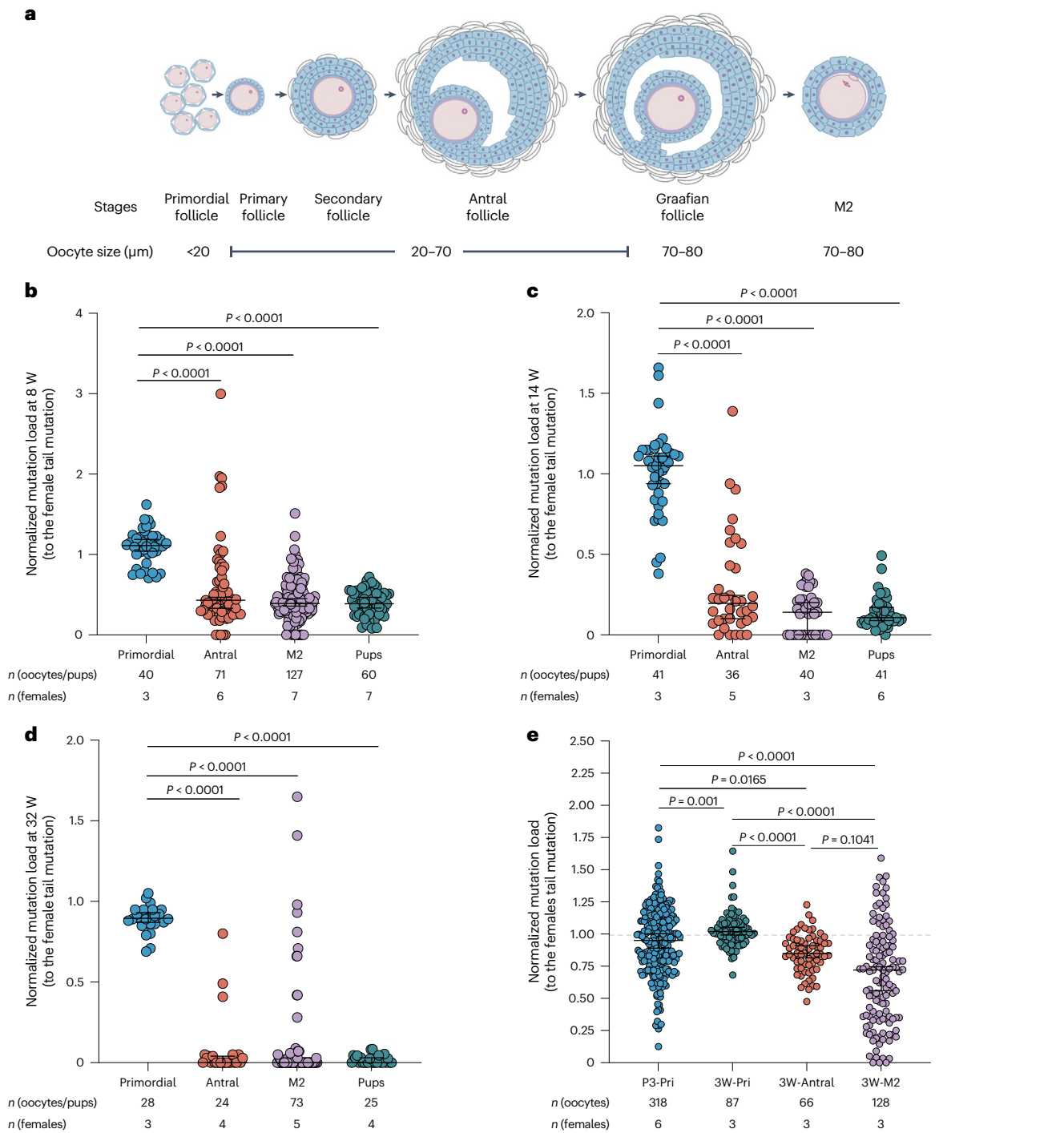


Fig. 4 | Age-enhanced purifying selection on m.2820 G > A occurs during oocyte maturation. a, Diagram of oocyte maturation within different ovarian follicles: oocyte (pink), granulosa cell (blue) and theca cell (white). Created with BioRender. **b**, Normalized mutation loads in individual oocytes from primordial and antral follicles, and at M2 stage, of 8-week-old female mice, alongside normalized mutation loads of pups born to females that became pregnant at 8 weeks of age. Every symbol represents an individual oocyte or pup. Data are median ± 95% CI. *P* values were determined by Kruskal–Wallis test followed by Dunn's multiple comparisons. **c**, As in **b**, for oocytes and pups from 14-week-old female mice. **d**, As in **b**, for oocytes and pups from 32-week-old female mice. **e**, Normalized mutation loads of individual oocytes at various stages from females less than 4 weeks of age, with 'Pri' indicating primordial follicle. Data are median ± 95% CI. *P* values were determined by Kruskal–Wallis test followed by Dunn's multiple comparisons. W, weeks.

male tail samples and did not exhibit any significant decline with age (Fig. 3g). Surprisingly, when analyzing the mutation load in individual sperm after flow cytometry sorting, a remarkable variation emerged, ranging from 0% to 100% (Fig. 3h). This indicates a significant heteroplasmic drifting during spermatogenesis, in which no selection occurred. Furthermore, it is noteworthy that the marked segregation of mtDNA during spermatogenesis did not affect the fertility of mutant males (Fig. 3i). High *ND1* mutant males did not show any changes in sperm count or sperm motility (Extended Data Fig. 3e,f). Therefore, it can be concluded that the age enhancement of purifying selection against the pathogenic *ND1* mutation is exclusive to the female germline.

Age-enhanced purifying selection occurs during oocyte maturation, not through oocyte atresia

The mammalian postnatal oocyte maturation process includes different follicle stages: primordial, primary, secondary, antral and Graafian follicles (Fig. 4a). Given the changes that follicles undergo during their maturation, we next sought to identify the precise stage of follicle development at which the age-enhanced selection phenomenon takes place. Additionally, we aimed to determine whether this phenomenon is mediated by oocyte atresia. To achieve this, we assessed the oocyte mutation load of follicles at different stages in females of different ages; the evaluated stages included primordial and antral follicles, which represent the starting point as the storage pool for all mature oocytes and the stage just before the completion of the first meiosis, respectively (Fig. 4a). Because isolating oocytes from primordial follicles poses a challenge, we conducted a benchmarking study to determine the diameters of these oocytes, as described in the Methods. Subsequently, we successfully isolated oocytes ranging from 14 µm to 20 µm, which we identified as oocytes from primordial follicles (Extended Data Fig. 4a,b). To our surprise, the mutation load was not decreased in oocytes from primordial follicles of 8-week-old, 14-week-old or 32-week-old mice. However, the mutation load of oocytes from the antral follicles of mice at all examined ages showed a very similar trend to those observed in M2 oocytes and pups (Fig. 4b–d), showing that the maternal age-enhanced selection event did not occur at the primordial follicle stage but, rather, at the following stages and was finished before completion of the first meiosis.

Because the oocytes from primordial follicles in mice shortly after birth serve as the reservoir for postnatal oocyte maturation for the whole life, we conducted further investigations on the mutation load in these oocytes. We observed relatively high mutation loads in oocytes derived from postnatal day 3 (P3) and P21 primordial follicles, as shown in Fig. 4e. It is worth mentioning that, although the 318 oocytes obtained from P3 mutant ovaries did not exhibit significant selection trends (with a median value of 0.9380, as shown in Fig. 4e), a closer analysis of the data revealed a slight increase in purifying selection when the female mutation load exceeded 37% but not when the mutation load was below 37% (Extended Data Fig. 5a). Similar trends were observed in females at P7 (Extended Data Fig. 5a). These findings indicate that purifying selection could occur during oogenesis in the embryonic stage and is closely associated with an individual's mutation load. Although these observations were made, it is important to note that maternal age-enhanced purifying selection plays a more prominent role in all females postnatally (with all data on oocytes or pups from individual females presented in Extended Data Fig. 5b–d), suggesting that maternal age-enhanced selection occurs primarily during postnatal oocyte maturation and is less sensitive to an individual female's mutation load.

Furthermore, we found that most ovulated oocytes in *ND1* mutants significantly decreased to 0% after 23 weeks of age, whereas we were unable to find a single oocyte without mutation among the 318 oocytes at P3 (Fig. 4e). Considering that there is no mitosis in postnatal oocyte maturation, we speculate that competition between oocytes with high and low mutations is not the main driver of selection, but, rather, selection at the organelle level is the only plausible mechanism. In other words, oocyte atresia, resulting from oocyte competition, is not the principal factor causing selection. Moreover, if oocyte atresia were the primary driving force, the strength of selection would vary significantly between females with differing mutation loads. High mutant females would face greater selection pressure, whereas low mutant females would experience weaker or no selection. However, our data demonstrate strong selection in all mutant females (Extended Data Fig. 5b–d).

To further confirm whether the *ND1* mutations could increase oocyte atresia, we conducted a systematic analysis of follicle numbers at various stages. Our analysis found no impact of the *ND1* mutation on follicle development at 14 weeks of age when maternal age-enhanced selection had already occurred since 8 weeks of age (Extended Data Fig. 5e,f). These results further support that follicle atresia is not the primary mechanism responsible for the purifying selection that occurs during postnatal oocyte maturation.

Taken together, these experiments demonstrate that age-enhanced purifying selection occurs during postnatal oocyte maturation, commencing after the primordial follicle stage and nearing completion before the completion of meiosis I.

Young females with high mutant oocytes can give rise to high mutant pups

There is a substantial selective pressure against the pathogenic *ND1* mutation during postnatal oocyte maturation after sexual maturity, resulting in a low level of mutant mtDNA (less than 50% of the mothers' tail mutation load) being passed on to the next generation. To investigate whether the high levels of mutant mtDNA present in mature oocytes can be transmitted to the next generation, we mated three groups of superovulated young mutants (the average tail mutation load of each donor group was approximately 37%, approximately 56% and approximately 69%) that were 3 weeks old. Zygotes from this mating were collected and transplanted into wild-type surrogate mothers using assisted reproductive technology (ART) (Fig. 5a; ART group). The obtained pups exhibited relatively high average mutation loads that were similar to the mutation loads in the donor mother tail samples. These results show that oocytes with high *ND1* mutation loads have the potential to develop into pups with similarly high *ND1* mutation loads (Fig. 5b). However, a mild purifying selection was observed in pups born to mothers with a high mutation load (~69%) (Fig. 5c), implying that there exists a limitation in the transmission of mutant *ND1* from mature oocytes to the next generation. Overall, these experiments demonstrate that high levels of mutations in mature oocytes can be inherited by subsequent generations, and ART can be used in 3-week-old females to maintain this mouse line at a high mutant level.

To investigate whether the upper limit of the mutation load in the pups could be further increased by a second round of mtDNA editing using DdCBE, we conducted editing on the remaining in vivo fertilized zygotes (as shown in Fig. 5a). As expected, this additional editing increased the average mutation load of the pups to 1.6-fold that of the donor females' group at approximately 37%. However, the effect of this additional editing on the zygotes from the donor females' group at around 69% was limited (Fig. 5d). In summary, our findings show that a second round of mtDNA editing can further increase the mutation load in the pups, with an upper limit of mutation load of 82.9%.

Age-enhanced purifying selection also acts on the *ND5* pathogenic mutation

To determine if a similar selection process occurs for other pathogenic mutations in the mitochondrial complex I genes, we generated a new mouse model with a mutation in the *ND5* gene. This gene encodes a subunit that is located far from the *ND1* subunit in the membrane part of complex I according to available structures[17,18]. We chose the 12918 site in mouse, which is homologous to the human 13513 site, known to cause MELAS syndrome in humans (Fig. 6a)[39–43]. This specific point mutation induces a substitution in the 393rd amino acid of the human and mouse ND5 protein, resulting in a conversion from aspartic acid (D) to asparagine (N). Using the same editing technique as for *ND1*, we obtained several single-cell-derived cell lines with varying levels of the *ND5* mutation, ranging from 1% to 94%. Our analysis of mitochondrial respiration with the XF Seahorse revealed that cell lines with high *ND5* mutation levels exhibited reduced OCRs (Fig. 6b). Next, we examined the MMP in these mutant MEFs, given that the targeted mutation site is located within a domain that is crucial in proton pumping activity by complex I. The MMP of the *ND5* mutant MEFs with 70–95% mutation load decreased to 75% of controls (to a much lesser extent than for the *ND1* mutant MEFs) (Fig. 6c).

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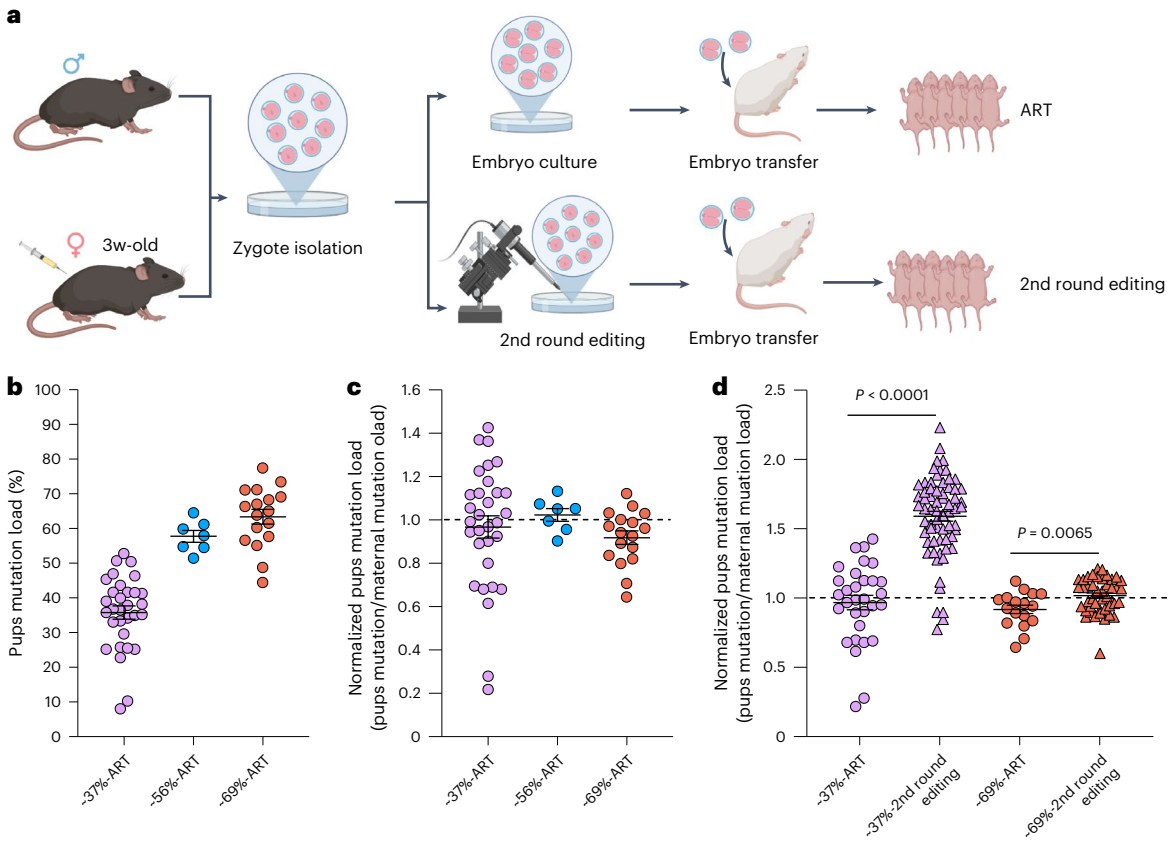


Fig. 5 | High *ND1* mutations in mature oocytes from young mice can be transmitted to the next generation. a, Experimental setup for ART and 2nd round of editing. Pups from both groups originated from an identical pool of zygotes. Created with BioRender. **b**, Mutation loads of pups born from ART-treated 3–4-week-old females of varying *ND1* mutation loads. Zygotes were collected from groups of females (>6 females per group) with average mutation loads of approximately 37%, 56% and 69%. Data are mean ± s.e.m. $n = 31$ pups in 37%-ART group, $n = 7$ pups in 56%-ART group and $n = 17$ pups in 69%-ART group. **c**, Normalized pup mutation loads as the same dataset from **b**. Data are mean ± s.e.m. **d**, Comparison of normalized pup mutation loads from ART and 2nd round editing procedures. Data are mean ± s.e.m. for $n = 31$ (37%-ART), $n = 67$ (37%-2nd editing), $n = 17$ (69%-ART) and $n = 45$ (69%-2nd editing) pups. *P* values were determined by two-tailed Mann–Whitney test. w, weeks.

Building on the data from MEFs, we performed editing in mouse zygotes and successfully obtained *ND5* mutant mice with an average mutation load of approximately 10.3% (range, 0–29%) (Fig. 6d). The editing efficiency for the *ND5* mutation in the obtained pups was lower compared to the efficiency achieved for the *ND1* mutation, where the average *ND1* mutation load of the pups was around 43% (range, 0–66.3%) at the 250 ng μl^{-1} concentration of injected mRNA. By backcrossing the *ND5* mutant females with B6 wild-type males, we observed age-enhanced purifying selection on the *ND5* mutation, as seen for the *ND1* mutation, although it was weaker (Fig. 6e and Extended Data Fig. 6a). The normalized mutation load of pups from *ND5* females aged 20–28 weeks was approximately 40%, which is equivalent to the mutation load of pups from *ND1* females aged 8 weeks. The weaker selection allowed us to maintain the mouse line with the *ND5* females aged 8 weeks, which produced pups at an almost neutral segregation pattern (Fig. 6e). With this, we could obtain *ND5* mutant mice with high mutation loads. Additionally, the litter size was not significantly different in the groups with high (39–51%) and low (12–16%) mutation loads compared to wild-type (Fig. 6f). A detailed assessment of the *ND5* mutation levels in M2 oocytes showed strong selection against the mutation, very similar to the selection against *ND1* in oocytes, but the onset of strong selection was at an older maternal age (*ND5* mice at 22–23 weeks of age; *ND1* mice at 8–9 weeks of age) (Figs. 3a and 6g and Extended Data Figs. 5c and 6b). The purifying selection against the *ND5* mutation was complete before the antral stage (Extended Data Fig. 6c). Still, this did not appear in the primordial follicle stage, regardless of maternal age (Extended Data Fig. 6c,d).

To evaluate the impact of potential nuclear off-target effects associated with the mtDNA editing technique on the maternal age-enhanced selection, we conducted a comprehensive analysis of selection intensity across various generations (F1, F2, F3, F5 and beyond F5) of backcrossing with B6 wild-type males. The consistent level of selection intensity observed in *ND5* mutant females across generations demonstrates that the selection does not stem from potential nuclear off-target effects. Rather, it can be attributed to the inherent pathogenicity of the point mutation itself during oocyte maturation (Fig. 6h).

Overall, our investigation extends the observations made regarding maternal age-enhanced selection on *ND1* mutations in oocytes to *ND5* mutations and consistently appeared across multiple generations.

Protein translation activity is increased in *ND1* mutant oocytes

The purifying selection of both *ND1* and *ND5* mutations was observed primarily in oocytes between the primordial and antral follicle stages, driven mainly by organelle-level selection rather than oocyte atresia (Figs. 4 and 6 and Extended Data Figs. 5 and 6). Here, we aimed to elucidate the molecular changes in these oocytes from *ND1* mutants to figure out the exact mechanism underlying purifying selection in these developing oocytes in the secondary follicles. Previous studies on oocytes focused primarily on the stages between the antral and M2 stages, as obtaining and evaluating oocytes at these stages is

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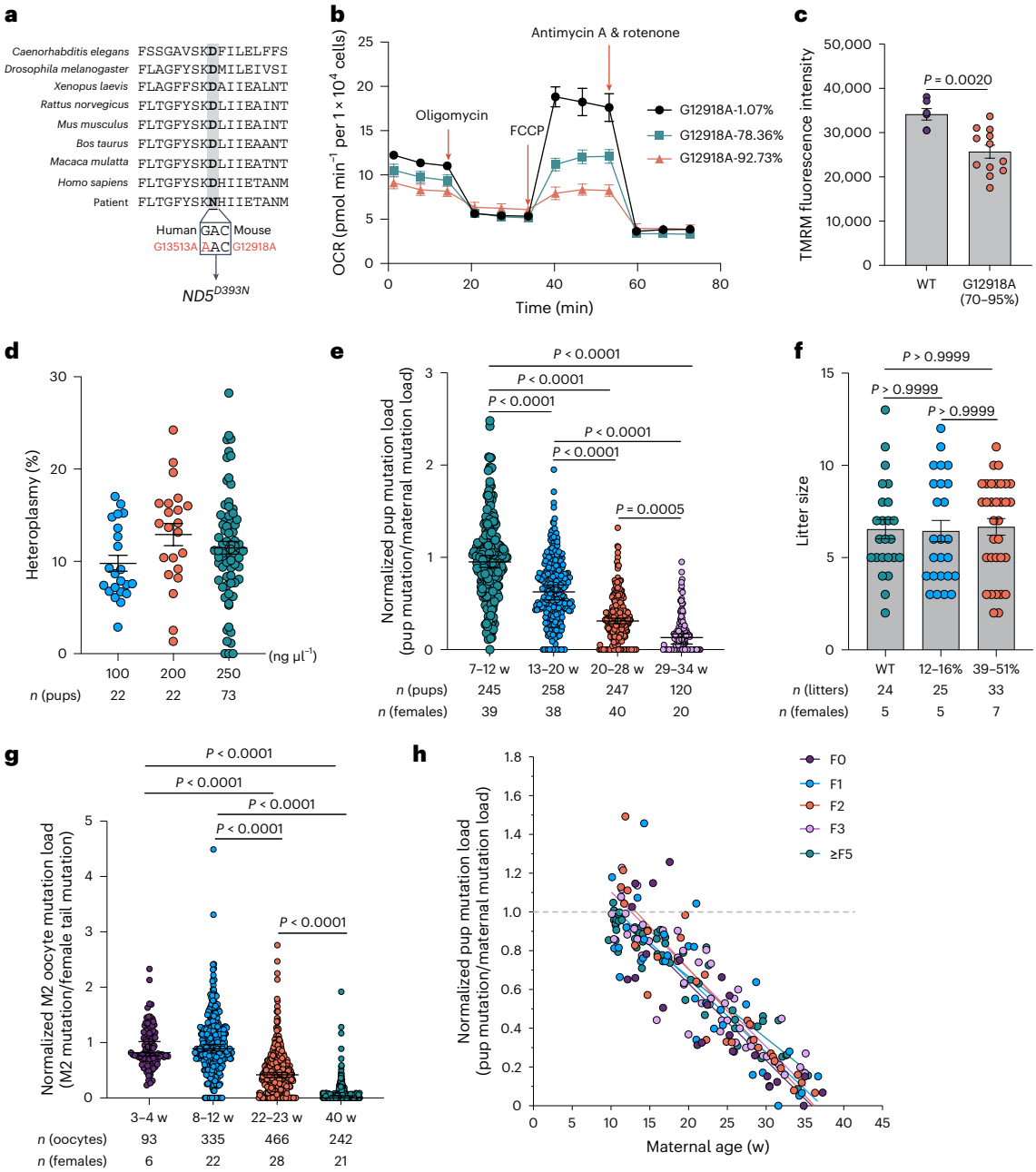


Fig. 6 | Age enhances purifying selection on the mouse *ND5* 12918 G > A mutation during transmission. a, ND5 protein sequence alignment (amino acids 385–401) highlights the evolutionary conservation at position 393 (bold, gray background). The human 13513 G > A mutation, analogous to the mouse 12918 G > A, predicts an aspartic acid (D) to asparagine (N) substitution in ND5. **b**, OCR in mutant MEF cells with varying m.12918 G > A mutation loads (1.07%, 78.36% and 92.73%) measured by Seahorse assay. Each group $n = 6$. Data are mean ± s.e.m. from ≥3 replicates. **c**, MMP in MEF cells indicated by TMRM fluorescence intensity. Data are mean ± s.e.m. $n = 6$ for WT; $n = 12$ for G12918A. *P* value was determined by two-tailed unpaired *t*-test. **d**, In vivo editing efficiency shown by mutation load in newborn pups. Injection mRNA concentration is on the *x* axis; pup mutation load is on the *y* axis. Each open circle represents a single pup. Data are mean ± s.e.m. **e**, Correlation between maternal age at conception and pup mutation load (pup toe mutation / maternal toe mutation). Each symbol represents one pup. Data are median ± 95% CI, with *P* value determined by Kruskal–Wallis test and Dunn's test. **f**, Breeding performance of *ND5* mutant females across mutation loads, compared to WT controls from Fig. 2f. Data are mean ± s.e.m., with *P* value determined by Kruskal–Wallis test and Dunn's test. **g**, Maternal age correlation with normalized mutation load in M2 oocytes from 77 mutant females. Each symbol represents an individual oocyte. Data are median ± 95% CI, with *P* value determined by Kruskal–Wallis test and Dunn's test. **h**, Trend of age-enhanced purifying selection across *ND5* mutant females at multiple generations (F0 to >F5) backcrossed with B6 WT males. Each symbol represents the average value of normalized pup mutation loads per litter. Total number of quantified litters: $n = 192$. w, weeks; WT, wild-type.

comparatively easier. Research on developing oocytes in secondary follicles has been relatively limited due to challenges associated with isolating oocytes at precise stages for comparison between mutants and controls. To precisely elucidate the effects of *ND1* mutations on gene expression in developing oocytes and the mechanisms underlying purifying selection, a cutting-edge technique called Stereo-seq was used[44]. This technique enables the detection of the transcriptome of ovarian sections in situ, with adjacent ovarian sections serving as morphological references for comparison of developing oocytes at the same stages.

Stereo-seq datasets were generated from four ovarian sections, two harboring the *ND1* mutation and two being wild-type, each originating from distinct mice (Fig. 7a). Adjacent tissue sections were stained with hematoxylin and eosin (H&E) to verify the presence and developmental stage of oocytes, and their spatial distribution on Stereo-seq slides was mapped. Through spatial expression pattern analysis of germ cell markers Gdf9 (refs. 45,46) and Dppa3 (also known as Stella or PGC7)[47], 17 candidate oocytes were identified across these tissues. Comparison with histological images from adjacent sections revealed the developmental stages of these oocytes: five in antral follicles, ten in secondary follicles (evenly distributed between the ND1 and wild-type groups), one in a primordial follicle and one in a follicle devoid of an oocyte (Extended Data Fig. 7a). Because selection occurs between the primordial and antral follicle stages, oocytes from secondary follicles were prioritized for differential gene expression analysis.

During differential gene expression analysis, two methodologies were implemented to ensure the reliability and validity of the results: the Wilcoxon rank-sum test and receiver operating characteristic (ROC) analysis. After the differentially expressed genes (DEGs) were identified, Gene Ontology (GO) enrichment analysis was performed. Common pathways were selected for further investigation based on the results obtained from both approaches (Extended Data Fig. 7b and Supplementary Tables 3–12). The shared GO pathway analyses revealed that the upregulation of pathways in *ND1* mutants, compared to wild-type controls, could be categorized into several groups, including 'Development' and 'Gene expression' (Fig. 7b–d). Further analysis revealed substantial increases in cytoplasmic translation and mitochondrial translation in the upregulated gene expression pathways of *ND1* mutants (Fig. 7e). These increases were particularly prominent in genes related to cytoplasmic ribosome subunits as well as mitochondrial ribosome subunits, suggesting an increase in translation within the oocyte driven by the *ND1* mutation (Fig. 7e,g). Conversely, the downregulated pathways related to 'Gene expression' in *ND1* mutants mainly involved RNA transcription and processing, albeit with milder trends (Fig. 7f,g). Additionally, upregulated apoptosis pathways were observed in the *ND1* mutants, indicating that the *ND1* mutations induce molecular changes related to apoptosis in the oocytes (Fig. 7h).

Because mTORC1 is a nutrient/energy/redox sensor that controls protein synthesis for cell growth[48–51], to investigate whether the upregulation of translation observed in the *ND1* mutant developing oocytes is mediated by the mTORC1 pathway, we conducted immunofluorescence staining for phosphorylated S6 (p-S6), a marker of mTORC1, on ovarian sections in situ. The results revealed a markedly stronger signal in *ND1* oocytes within both secondary and antral follicles when compared to the wild-type control (Fig. 8a,b). This finding reveals that there is indeed an enhanced translational activity as a result of the *ND1* mutation.

In summary, our Stero-seq analysis demonstrated a significant increase in translation in *ND1* oocytes. This elevation in translation may be regulated by the mTORC1 pathway in *ND1* females.

Increased translation activity plays an important role in the purifying selection process

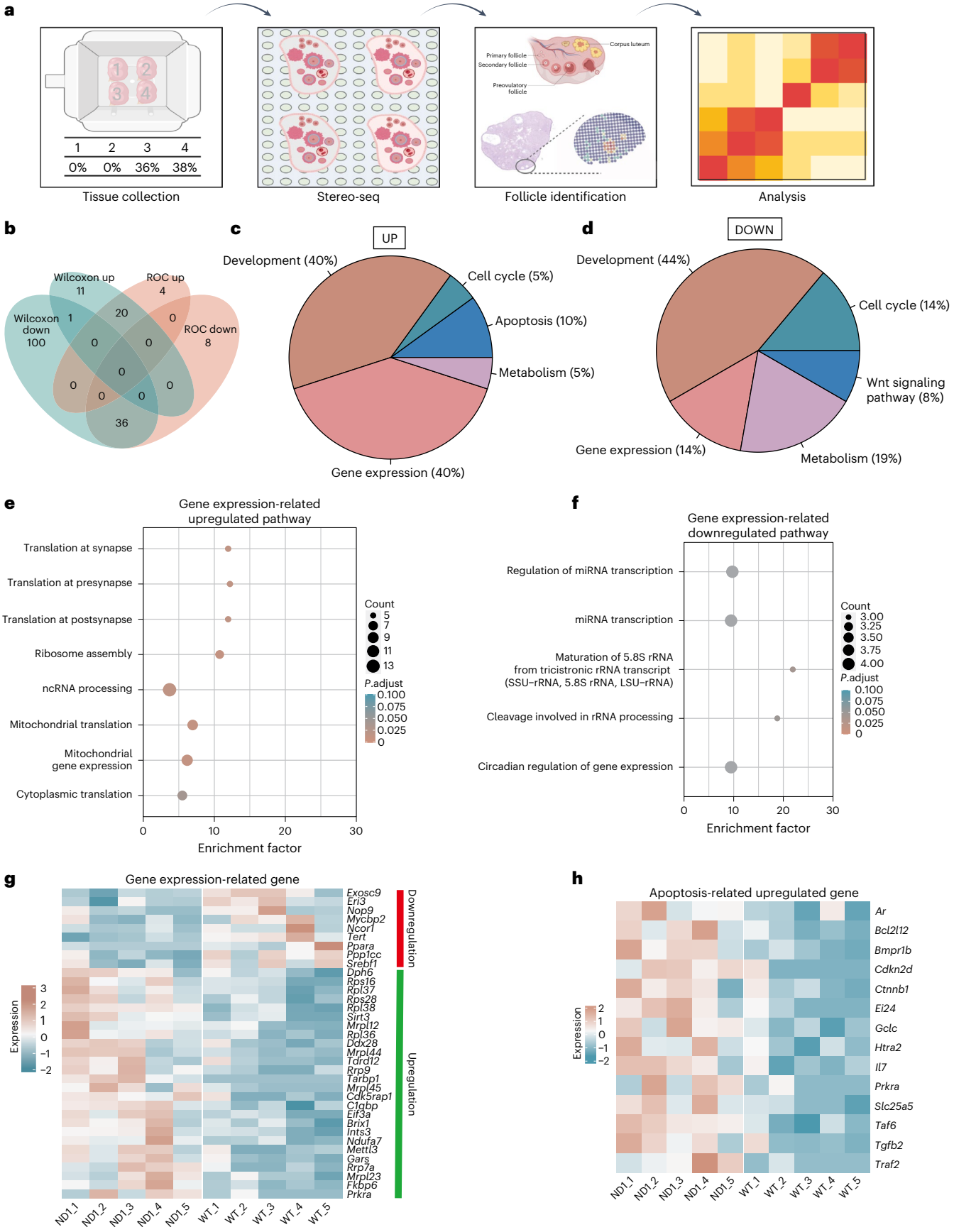
Given the observed increase in protein translation in our *ND1* mutant oocytes, we hypothesize that this event may play a role in purifying selection on *ND1* mutations during oocyte development. To test this hypothesis, we conducted rapamycin intraperitoneal injection experiments in mice to suppress mTORC1 and inhibit translational activity in oocytes. We used both long-term (49 d, spanning from primordial follicle to antral follicle stage) and short-term (21 d, spanning from partial secondary to antral follicle stage)[52] treatments and detected oocyte mutation loads from treated females at 17 weeks of age (Fig. 8c). Both treatments could reduce p-S6 levels in the heart and ovaries, as detected by western blot (Extended Data Fig. 8a–f), confirming rapamycin's efficacy in inhibiting protein translation in vivo. Notably, although mouse body weight remained stable (Fig. 8d), a decrease in ovarian weight without affecting the number of M2 oocytes after superovulation was observed (Fig. 8e,f), consistent with previous studies[53]. Immunofluorescence analysis of ovarian cryosections from rapamycin-treated mice revealed a marked reduction in p-S6 signals in the oocytes from *ND1* mutant ovarian sections, indicating suppressed translational activity (Fig. 8g,h). Interestingly, the detailed analysis of the *ND1* mutation load in these oocytes revealed a significant reduction in the strength of purifying selection in M2 oocytes in the long-term rapamycin treatment group, but not in the short-term treatment group, compared to the PBS treatment group (Fig. 8i). To further evaluate the inhibitory strength, we compared the mutation load of all the M2 oocytes from untreated females aged 8–9 weeks and 14 weeks, previously measured (Fig. 3a), to 17-week-old females with treatment (Fig. 8i). We found that long-term rapamycin treatment completely blocked purifying selection when compared to the 8–9-week-old untreated group (Extended Data Fig. 8h). The strong inhibitory effect suggests that inhibiting translation in oocytes, particularly from the primordial to antral stages, disrupts the natural selection process during oocyte maturation, highlighting the critical role of enhanced protein translation in the purifying selection of mtDNA mutations. Furthermore, we also found that the PBS treatment group did not show a significant difference compared to the 14-week-old untreated group (Extended Data Fig. 8h; PBS treatment versus 14-week M2), suggesting that a 3-week interval may be insufficient for detecting selection effects in females aged from 14 weeks, which could potentially explain the absence of an inhibitory effect in the short-term rapamycin treatment starting from 14 weeks. Notably, although rapamycin substantially reduced protein translation activity in the heart, it did not alter the heart's mutation load (Extended Data Fig. 8g). This difference further supports the notion that our findings are specifically related to the oocyte maturation process.

Due to the mitophagy-promoting function of rapamycin and its inhibitory effect on observed purifying selection[54], we further hypothesize that mitophagy may not be the cause of selection in developing oocytes with *ND1* mutations. This hypothesis contradicts previous studies that suggest the role of programmed germline mitophagy (PGM) in promoting purifying selection in *Drosophila* germline[27]. To confirm this hypothesis, we evaluated the level of mitophagy in the developing oocytes of *ND1* mutant and wild-type mice to investigate whether mitophagy occurs and helps clear *ND1* mutations. First, we analyzed Stereo-seq data and found no significant differences in the expression of genes related to mitophagy between wild-type and *ND1* mutant oocytes (Extended Data Fig. 9a). Moreover, electron microscopy analysis revealed the absence of typical structures of mitochondrial autophagy in the developing oocytes from both groups. Although some autophagosome-like structures were observed, there

Fig. 7 | Enhanced protein synthesis in oocytes from *ND1* mutant females elucidated by spatial transcriptomics. a, Fresh ovarian tissue was embedded in OCT and sectioned, and high-quality slices were placed on a spatial transcriptomics chip for sequencing. Adjacent sections underwent H&E staining for histological analysis. Germ cell markers and morphology aided in identifying oocytes within secondary follicles for subsequent data analysis. Created with BioRender. **b**, Venn diagram displaying overlapping pathways between two analyses, using a *q* value of 0.05 for significance determination. **c**, Pie chart summarizing upregulated pathways shared by the two analyses in **b**. **d**, Pie chart summarizing the downregulated pathways shared by the two analyses in **b**. **e**, Detailed GO pathways in the upregulated 'Gene expression' category in secondary follicle oocytes from *ND1* mutants versus WT controls. **f**, Detailed GO pathways in the downregulated 'Gene expression' category in secondary follicle oocytes from *ND1* mutants versus WT controls. **g**, Heatmap presenting detailed gene expression profiles in the upregulated and downregulated 'Gene expression' category, log normalized for clarity using Seurat. **h**, Heatmap presenting detailed gene expression profiles in the upregulated apoptosis pathway, log normalized for clarity using Seurat.

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were no significant differences between the two groups (Extended Data Fig. 9b,c). Subsequently, we conducted immunofluorescence experiments to examine the co-localization of autophagy-related marker proteins, specifically LC3B and LAMP1, and mitochondrial marker proteins, such as MTCO1 and TOMM20, in these developing oocytes. We did not observe any co-localization of mitochondrial markers with



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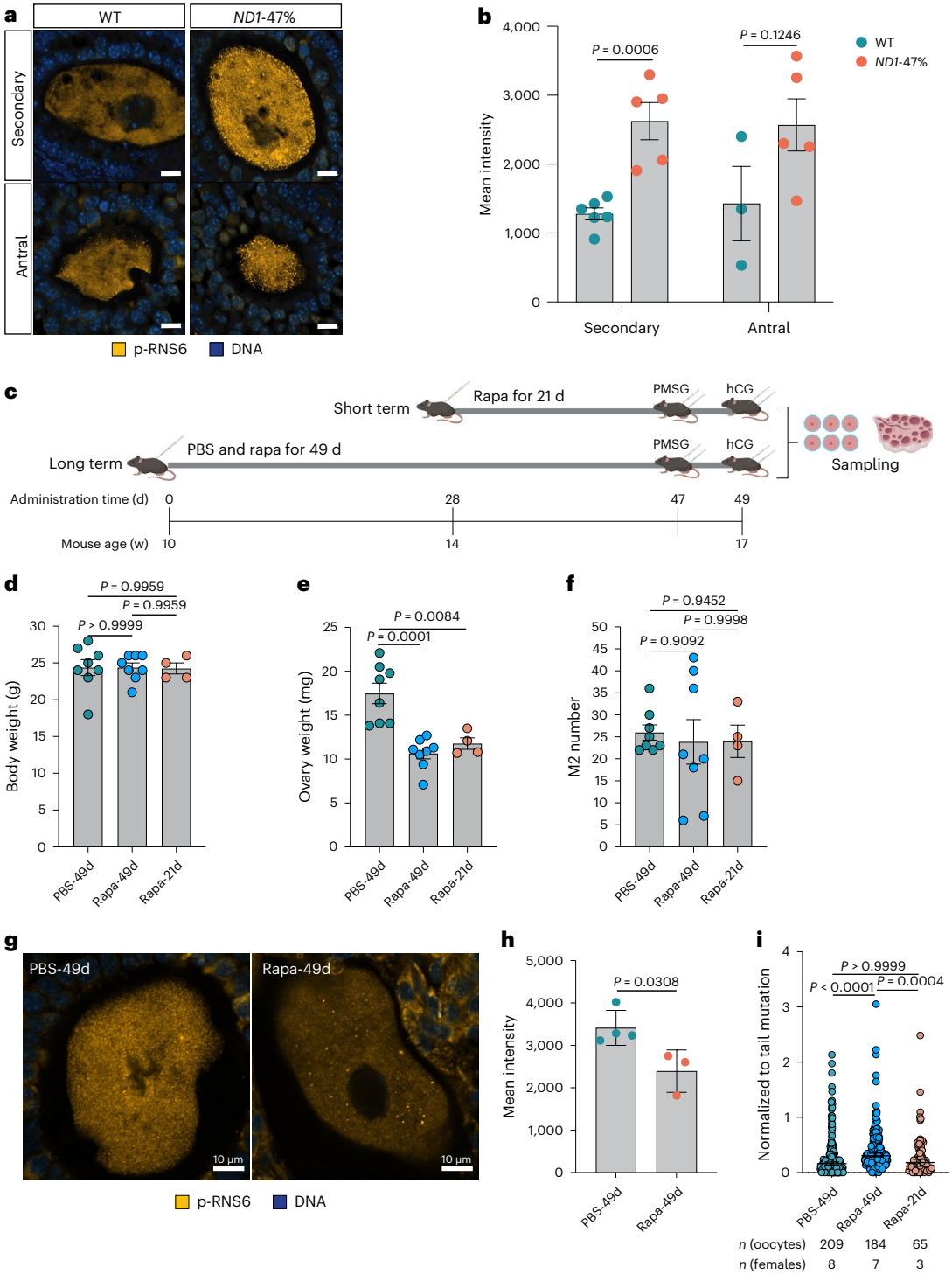


Fig. 8 | Enhancement of protein translation drives purifying selection in *ND1* mutant oocytes. a, Representative images of oocyte p-S6 immunofluorescence within secondary and antral follicles, indicative of protein translation activity. Scale bar, 10 µm. **b**, Quantification of p-S6 fluorescence intensity in secondary and antral follicle oocytes from *ND1* mutants (secondary: $n = 5$; antral: $n = 5$) and WT controls (secondary: $n = 6$; antral: $n = 3$). Data are mean ± s.e.m., with *P* values determined by two-tailed unpaired *t*-tests. **c**, Treatment and analysis timeline: rapamycin or PBS administration, hormone induction and assays. *ND1* mutant mice at 14 weeks received 21-d rapamycin (short term), whereas 10-week-old mice received 49-d treatment (long term), with all mice euthanized at 17 weeks. Created with BioRender. **d**–**f**, Mouse metrics after rapamycin or PBS treatment: body weight (**d**), ovarian weight (**e**) and M2 oocyte count (**f**). PBS-49d group: $n = 8$; Rapa-49d group: $n = 8$; Rapa-21d group: $n = 4$. Data are mean ± s.e.m. *P* values were determined by two-tailed unpaired *t*-tests. **g**, Representative images of p-S6 immunofluorescence in antral follicle oocytes from mice treated with rapamycin or PBS. **h**, Quantification of p-S6 fluorescence intensity in antral follicle oocytes. Sample sizes: PBS-49d, $n = 4$; Rapa-49d, $n = 3$. Data are mean ± s.e.m. *P* values were determined by two-tailed unpaired *t*-tests. **i**, Normalized mutational loads in M2 stage oocytes from treated mice. Each symbol represents an individual oocyte. Median ± 95% CI is presented, with *P* values determined by Kruskal–Wallis test and Dunn's post hoc analysis. Rapa, rapamycin; w, weeks.

autophagy-related proteins (Extended Data Fig. 9d,e). Furthermore, LC3B signals were not even detectable in developing oocytes from secondary follicles, which is consistent with previous reports[55]. These findings collectively indicate the absence of observable mitophagy in the *ND1* mutant developing oocytes and support the notion that mitophagy does not play a role in clearing *ND1* mutations.

In summary, the primary factor driving the purifying selection of *ND1* mutant mtDNA in developing oocytes is the increased protein translation activity, with limited supportive evidence for mitophagy.

Discussion

Our study demonstrated that strong purifying selection shapes the transmission pattern of two mtDNA mutant complex I gene mouse models with homologous mutations causing human mitochondrial diseases. Moreover, we found that maternal age further enhances the selection process. We also determined that this selection occurs during postnatal oocyte maturation, with increased protein synthesis playing a prominent role in this process. These findings provide key insights into understanding the transmission pattern of human mitochondrial diseases and the impact of maternal age on the selection and prevention of pathogenic mutations in mitochondrial diseases.

The strong purifying selection observed in our mouse models helps explain the rare reporting and low prevalence of these two homologous mutants, *ND1* (3376 G > A) and *ND5* (13513 G > A), in human patients[33,39–43]. This selection is also evident in the pathogenic *Cox1* mutants in *Drosophila* and *ND6* mutants in mice, suggesting its universality for certain types of mtDNA mutations, such as mitochondrial mRNAs. Moreover, our findings with the *ND1* and *ND5* mutations, along with the previously reported *ND6* mutation, demonstrate that purifying selection takes place in three of the seven mitochondrial genes encoding complex I subunits when mutated during transmission. This indicates that such a selection pattern could be a common phenomenon for mtDNA mutations in complex I genes.

A particularly noteworthy finding from our study is that the selection process occurs in an age-enhanced manner during transmission, with varying strength depending on the mutation. The faster selection speed in our *ND1* mouse models, compared to *ND5* models, aligns with the relatively lower prevalence of human mitochondrial diseases caused by *ND1* (3376 G > A) mutations compared to *ND5* (13513 G > A) mutations. This suggests two things. First, maternal age may act as a protective factor for inhibiting certain pathogenic mtDNA mutations to be transmitted, contradicting the conventional notion that maternal age is always a negative factor for passing on mutations to future generations. Second, the age-enhanced strength of selection is mutation dependent, further shaping the transmission patterns caused by different pathogenic mutations in a fine-tuned manner. The observed effect of maternal age on the transmission of mtDNA is consistent with a previous study conducted on healthy human multigenerational families, which demonstrated an increasing divergence between maternal and filial mtDNA as delivery age increases[11]. Based on our findings, we recommend including maternal age at pregnancy in future clinical retrospective studies of mitochondrial diseases. Collecting more comprehensive clinical data will help better understand these transmission patterns.

After determining the mutation type as well as the maternal age enhancement for this purifying selection, we delved further into the developmental stage through which oocytes execute this selection process. We observed that the age-enhanced purifying selection persisted for over 8 months in our *ND* mouse models. Considering the varying lifespans of different follicles, we hypothesize that long-lived oocytes within primordial follicles likely rely on functional complex I to maintain basic metabolic functions and could accumulate damage from defective complex I. This would enable developing oocytes of different ages to detect varying levels of damage and initiate a more robust selection process with increasing age. Our hypothesis contradicts a recent study that demonstrated that complex I is neither assembled nor active in early oocytes of humans and *Xenopus*[56]. If correct, purifying selection on mutations in complex I genes would not present in an age-enhanced manner, as all developing follicles originate from the same oocyte pool. Further research is needed to establish whether species differences account for this inconsistency. Additionally, we found that the mutation load remained constant in all tested oocytes in primordial follicles at any detected age point, implying that this cell population remains predominantly quiescent with minimal mitochondrial turnover, including mtDNA replication. Consequently, accumulated mitochondrial damage in these oocytes would only initiate selection in subsequent developing stages, characterized by a marked increase in mitochondrial biogenesis and mtDNA replication[57,58], as evidenced by our mutation load analysis of oocytes from various stages. Interestingly, the selection stage for our *ND* mouse models occurs much later than the germline 16-cell cyst stage in *Drosophila Cox1* mutant models (equivalent to the germ cell cysts during mammalian embryonic development), where selection of mutations in complex IV genes takes place. The differences in pathogenicity and the specific complexes containing these mutations (complex I versus complex IV), coupled with the distinct oogenesis processes in mammals and *Drosophila*, could account for this difference.

The purifying selection process has been the subject of ongoing debate, focusing on whether it occurs primarily at the cellular level, at the organelle level or even at the level of mtDNA. Specifically, the question remains unanswered as to whether the selection process at the organelle level involves the preferred biogenesis of healthy mitochondria or through mitochondrial quality control mechanisms, such as mitophagy. In the final part of our study, we established that the observed selection takes place predominantly at the organelle level, specifically during oocyte maturation, rather than through cellular competition, known as oocyte atresia. Through the use of Stereo-seq, long-term rapamycin treatment and mitophagy detection in developing oocytes, we elucidated that *ND1* mutant oocytes undergo upregulated protein translation to facilitate the purifying selection of these mutants. Notably, the administration of rapamycin, known for its ability to promote mitophagy, weakened the strength of the selection, suggesting that mitophagy may not contribute to the process. Notably, our results did not reveal the occurrence of mitophagy in mouse developing oocytes, including both wild-type and *ND1* mutants. This finding differs from previous reports of PGM in the *Drosophila* female germline. This discrepancy may be attributed to the different stages observed in our study compared to previous *Drosophila* studies. Our study focused on developing oocytes during postnatal oocyte maturation, whereas PGM predominantly occurs during the stage equivalent to mammalian embryonic oogenesis. Overall, our data strongly support the notion that the purifying selection heavily depends on protein translation and cell growth pathways, indicating that enhancing the generation of healthy mitochondria may be the driving force behind our selection process. However, further research is necessary to fully comprehend how *ND1* mutation increases translation levels and its detailed contribution to the proliferative advantage of healthy mitochondria.

In summary, our study offers, to our knowledge, the first solid evidence that maternal age can have a beneficial effect in preventing the transmission of pathogenic mtDNA mutations, particularly in genes encoding mitochondrial complex I subunits. Our findings emphasize the importance of examining the relationship between the age of pregnant people with mtDNA mutations and the mutation load in their children. Additionally, our research provides valuable strategies for creating and maintaining *ND* mouse models with high mutation loads, facilitating the future study of the pathogenesis of mitochondrial diseases.

Methods

Construction of *ND1* and *ND5* editing plasmids with fluorescent reporters

To evaluate the editing efficiency of DdCBEs, we engineered plasmid pairs for each target site, replacing the puromycin resistance cassette from the original constructs[13,59] with fluorescent reporter genes (mCherry or eGFP). This substitution facilitates transfection efficiency monitoring and the application of FACS. The optimized DdCBE systems include mitochondrial target sequence (MTS)-transcription activator-like effector (TALE) split halves of the DddA deaminase (split at either G1333 or G1397 of DddA)-UGI and mCherry or eGFP reporters (Extended Data Fig. 1).

We assembled four plasmid combinations for each mtDNA site (L_G1333N/R_G1333C, L_G1333C/R_G1333N, L_G1397N/R_G1397C and L_G1397C/R_G1397N) to determine the most efficient pairing for editing. We quantified editing efficiency by analyzing mutation load in the target mtDNA regions. TALE sequences targeting specific mtDNA sites were synthesized by Suzhou GENEWIZ Biotechnology Co. These sequences were cloned into our vectors using a ClonExpress Ultra One Step Cloning Kit (C115, Vazyme) by homologous recombination. The plasmid sequences for *ND1* and *ND5* MEFs and mouse models with the highest efficiency are listed in the Supplementary Information.

MEF cell culture and mtDNA editing

MEF cells (SCRC-1045, American Type Culture Collection) were cultured at 37 °C and 5% CO_2 in DMEM (11965092, Gibco), supplemented with 10% v/v FBS (12664025, Gibco) and 1% penicillin–streptomycin (GENOM). For mtDNA editing, 4 μg of DdCBE plasmid pairs was introduced into 5×10^5 MEF cells by nucleofection (4D-Nucleofector, Lonza) set to the 'MEF cell line' mode. Nucleofected cells were seeded into six-well plates and incubated for 3 d. Cells were then trypsinized (0.25%, GENOM) and sorted using FACS (channels: AF488 and mCherry) with a MoFlo Astrios cell sorter (Beckman Coulter). eGFP/mCherry double-positive cells were expanded for six additional days before Sanger sequencing analysis to determine the plasmid pair with the highest editing efficiency.

After identification, MEF cells were transfected with the most efficient plasmid pair to generate single-cell-derived clones for further biochemical and functional analysis. eGFP/mCherry double-positive single cells were isolated and cultured in 96-well plates. After expansion (10–15 d), colonies were divided and partially sent for mutation load assessment using Hi-TOM, and the remainder was maintained for cryopreservation and additional experimentation.

Quantification of mtDNA mutation load

Mutation load was measured using two methods. Sanger sequencing provided an initial assessment of DdCBE editing in MEF cell lines (Fig. 1b). Hi-TOM[34], a next-generation sequencing (NGS)-based method, was used for precise mutation quantification across multiple samples. For accuracy validation, we created two plasmids (pBM16A Topo smart kit, CL071, Biomed) with wild-type and G2820A mutant mtDNA fragments amplified from mutant MEFs to simulate different mtDNA contexts. Mixing these plasmids at various ratios generated standards representing mutation loads from 0% to 100%. Hi-TOM sequencing of these standards resulted in a highly accurate mutation load measurement ($R^2 = 0.9983$; Extended Data Fig. 1d). Primers used for mutation load detection are listed in the Supplementary Information.

After validation, Hi-TOM sequencing was used to measure mutation loads in mutant MEFs, mice and their tissues. Mutation load in each mouse was estimated using DNA from toe clips at 1–2 weeks of age or from tail clips at the time of euthanization.

Mitochondrial respiration measurements

Mitochondrial OCRs in MEFs were measured using a Seahorse XFe 96 Extracellular Flux Analyzer (Seahorse Bioscience). MEFs were seeded at a density of 18,000 cells per well in XFe 96-well microplates pre-coated with polylysine (10 mg ml^{-1}) and cultured in high-glucose DMEM for 16 h. Before analysis, the medium was exchanged to 175 μl per well of XF assay medium (XF Base Medium supplemented with 10 mM glucose, 2 mM L-glutamine and 2 mM sodium pyruvate, pH 7.4), and cells were incubated at 37 °C for 1 h for balancing. The Seahorse sensor cartridge was loaded with mitochondrial inhibitors: oligomycin (final concentration: 2 μM), FCCP (0.75 μM), rotenone (1 μM) and antimycin A (1 μM). The assay protocol consisted of 3-min mixing, 0-min wait and 3-min measurement cycles, repeated five times. After assay, cell numbers were quantified using a CyQUANT Cell Proliferation Assay kit (V13154, QIAGEN). OCRs were normalized to cell number and analyzed using Wave software (version 2.6, Agilent Technologies).

BN-PAGE and in-gel activity assays

Mitochondria were isolated from MEFs using a modified protocol[60]. MEFs cultured to confluence were harvested and subjected to osmotic lysis in RSB Hypo buffer (10 mM NaCl, 1.5 mM $MgCl_2$ in 10 mM Tris-HCl, pH 7.5). Subsequent homogenization was performed with a rotating Teflon homogenizer (1,600 rpm, 60 strokes). Homogenization was checked visually under a phase-contrast microscope (Nikon Ts2). When the lysis degree (8–9 naked nuclei for every whole cell) of homogenization was achieved, the homogenate was balanced with 2.5× MS homogenization buffer to 1× (210 mM mannitol, 70 mM sucrose, 1 mM EDTA in 5 mM Tris-HCl buffer, pH 7.5), followed by differential centrifugation (1,000*g* for 10 min, followed by 11,000*g* for 15 min). The mitochondrial pellet was resuspended in 1× MS homogenization buffer, and protein concentrations were determined using the Bradford assay.

For BN-PAGE, 400 μg of isolated mitochondria was lysed in 10 μl solubilization buffer (100 mM Bis-Tris (pH 7.5), 500 mM ε-amino-*n*-caproic acid, 50 mM NaCl, 1 mM EDTA (pH 7.5), 1 mM PMSF, 10% (v/v) glycerol) containing 240 μg μl^{-1} digitonin (D141, Sigma-Aldrich) and mixed with loading dye (5% (w/v) Coomassie Brilliant Blue G-250, 100 mM Bis-Tris (pH 7.5) and 500 mM ε-amino-*n*-caproic acid) and resolved in 4–10% gels, which were then either subjected to Coomassie staining or processed for in-gel activity assays.

The in-gel activity of complex I (NADH reductase capacity) was determined by incubating the native gels in complex I assay solution (5 mM Tris-HCl, pH 7.4, 2.5 mg ml^{-1} nitrotetrazolium blue (298-83-9, Sigma-Aldrich) and 0.1 mg ml^{-1}NADH (606-68-8, Roche)) for 30 min at room temperature. The in-gel activity of complex IV was determined by incubating the native gels in complex IV assay solution (50 mM phosphate, pH 7.5, 0.5 mg ml^{-1} DAB and 1 mg ml^{-1} cytochrome c) for 1–2 h at room temperature. Both assays were terminated with 10% acetic acid.

Western blot

MEF cell pellets were lysed in RIPA buffer (P0013B, Beyotime) containing a protease inhibitor cocktail (A32963, Roche) and PMSF (8553S, Cell Signaling Technology), and protein quantification was performed using a BCA kit (C503021, Sangon). Proteins were then resolved by SDS–PAGE (4–12% gradient gels) and transferred to PVDF membranes. Membranes were blocked with 5% non-fat milk in TBST buffer for 1 h at room temperature, followed by primary antibody incubation overnight at 4 °C. HRP-conjugated secondary antibodies were applied for 1 h at room temperature. Immobilon Western Chemiluminescent HRP Substrate (Merck) was added according to the manufacturer's instructions, and the protein bands were visualized and recorded with an Amersham Imager 680 system (GE). Densitometry was performed using FIJI software and normalized to internal controls. Antiobodies used here are listed in the Supplementary Information.

Maintenance of mice and animal ethics

All procedures involving animals were conducted in strict accordance with the guidelines for the Care and Use of Laboratory Animals of Westlake University and received approval from the Institutional

Animal Care and Use Committee at Westlake University (approval no. 22-003-JM-3). The mice were housed under a 12-h light–dark cycle at controlled ambient temperature (22 ± 1 °C) and relative humidity (40–70%). Adequate standard chow and water were available ad libitum via cage lids. Detailed information on the animals' age and sex is specified within the main text and associated figure legends.

For the maintenance of the *ND1* and *ND5* mutant mouse lines, breeding with wild-type C57BL/6J male mice was performed. Individual *ND1* or *ND5* mutant female mice (8–10 weeks old) with varying mutation loads was paired with male mice (8–10 weeks old) for fertility assessments over a 6-month period. This breeding setup was also employed to evaluate fertility in male *ND1* mutant mice. Litter sizes and birth dates were recorded for analysis.

Generation of *ND1* and *ND5* mutant mouse models

DdCBE editing plasmids with the highest editing efficiency in MEF cells were linearized with Xho I (New England Biolabs) and purified with an AxyPrep PCR Clean-up Kit (AP-PCR-250). Linearized plasmids were then transcribed in vitro with an mMESSAGE mMACHINE T7 Ultra Kit (A57620, Life Technologies) according to the manufacturer's instructions. This kit includes an anti-reverse cap analog (ARCA), ensuring that T7 RNA polymerase initiates transcription with the correct orientation, resulting in the production of RNA molecules that are properly capped. After transcription, the synthesized RNA was purified with a MEGAclear Kit (AM1908, Life Technologies), following the manufacturer's manual. The integrity of the purified RNA was verified through denaturing agarose gel electrophoresis, and samples were stored at −80 °C for future injection.

Zygote microinjection involved superovulating 3-week-old C57BL/6J female mice with intraperitoneal injections of 10 IU of pregnant mare serum gonadotropin (PMSG, Ningbo Second Hormone Factory), followed by 10 IU of human chorionic gonadotropin (hCG, Ningbo Second Hormone Factory) 48 h later. These superovulated females were then paired with C57BL/6J males (aged 8–12 weeks) for mating. Zygotes were collected from the oviducts 16 h after hCG administration. Subsequently, the DdCBE mRNA was microinjected into the cytoplasm of the collected zygotes.

For embryo transfer, ICR pseudo-pregnant female mice (8–10 weeks old) at 0.5 days postcoitum, having mated with vasectomized males, were anesthetized with avertin (240 mg kg^{-1} body weight). The microinjected zygotes were then transferred into the oviducts using a fine glass capillary.

For producing pups from young *ND1* mutant females (aged 3–4 weeks), the superovulation protocol was repeated as above, and, after mating with wild-type C57BL/6J males, the zygotes were collected. These were allocated into two groups: the ART group, wherein zygotes were directly transferred to the oviducts of pseudo-pregnant ICR mice, and the second-round-editing group, wherein zygotes were subjected to another round of DdCBE mRNA microinjection (at a concentration of 250 ng μl^{-1}) before transfer to the ICR recipients.

Mitochondrial genome-wide off-target assay

To comprehensively assess the mtDNA off-target effects of DdCBE in our mutant mouse model, we isolated mitochondria from mutant mouse hepatic tissues following a previously reported protocol[60]. In brief, liver samples were cut into pieces. These tissues were then thoroughly washed and homogenized in 1× MS homogenization buffer until only connective tissue was discernible as introduced above. After homogenization, the sample was subjected to a two-step centrifugation protocol (1,300*g* for 10 min and collecting supernatant at 11,000*g* for 15 min).

The obtained mitochondrial pellet was then processed for DNA extraction using a FastPure Cell/Tissue DNA Isolation Mini Kit (DC102-01, Vazyme), following the manufacturer's instructions. The purified mtDNA was then subjected to NGS.

Isolation of oocytes at various developmental stages

Our approach to isolating primordial follicle oocytes involved benchmarking oocyte diameters using consistent dissection microscopy. Oocytes were isolated from the ovaries of P3 mice, which contain exclusively primordial follicles and somatic cells at this stage. Measurement of a large number of oocyte diameters indicated that most oocytes ranged from 12 µm to 22.1 µm, with a mean diameter of 15.3 µm (±2.058 µm), which is in agreement with previous studies[61] (Extended Data Fig. 4a,b). These diameters were significantly larger than those of somatic cells (4.8–11.4 µm, mean diameter of 8.1 µm (±1.446 µm)). Consequently, a standard diameter range of 14–20 µm was established for the isolation of primordial follicle oocytes in adult ovaries.

For the isolation of primordial follicle oocytes from P3 and P7 mice, ovaries were held with tweezers and gently scraped in a Petri dish containing MEMα to release the cells. For mice at stage P21 and older, the isolated ovaries were cut into approximately 1-mm^3 blocks in MEMα. Cells with diameters between 14 µm and 20 µm were identified as primordial follicle oocytes and were individually collected and transferred to PBS for washing.

Antral oocytes from mice at P21 and older were retrieved from the remaining suspension after primordial follicle oocyte selection. Cumulus cells were removed by pipetting under a dissecting microscope, and antral oocytes were washed in PBS and collected individually.

For M2 oocyte collection, female mice were superovulated using the previously described protocol: 10 IU PMSG followed by 10 IU hCG. The mice were euthanized for oviduct dissection. Cumulus–oocyte complexes (COCs) were released from the ampulla and denuded with 0.6 mg ml^{-1} hyaluronidase (H3506, Sigma-Aldrich) at 37 °C for 5 min. M2 oocytes were washed three times in M2 medium (M7167, Sigma-Aldrich) and collected individually in PBS.

To extract DNA from oocytes for mtDNA mutation load measurement, lysis buffer (0.5% Tween 20, 100 µg ml^{-1} proteinase K in 50 mM Tris-HCl buffer, pH 8.5) was used. The lysis conditions varied: 37 °C for 30 min for primordial follicle oocytes and 55 °C for 16 h for antral and M2 oocytes, followed by inactivation at 95 °C for 10 min. The DNA was then amplified by PCR and analyzed by Hi-TOM.

Analysis of mouse sperm motility and mutation load

For the evaluation of sperm motility, the caudal epididymis of male mice was dissected, cleared of surrounding tissue and placed in a 1.5-ml EP tube containing pre-warmed Biggers, Whitten and Whittingham (BWW) medium (94.8 mM NaCl, 4.8 mM KCl, 1.2 mM $MgSO_4$, 1.2 mM KH_2PO_4, 1.7 mM $CaCl_2$, 25 mM $NaHCO_3$, 13.21 mM sodium lactate, 0.27 mM pyruvate, 5.5 mM glucose, 10 mM HEPES and 2.6% BSA, pH 7.2). The epididymis was cut to release spermatozoa into the medium, which was then analyzed using computer-aided sperm analysis (CASA) with an IVOS sperm motility analyzer (V12.3A, Animal Motility, Hamilton Thorne Research) to determine concentration and motility parameters.

To determine mutation loads in bulk sperm, sperm pellets were lysed with the lysis buffer (1 M Tris-HCl, pH 7.5, 0.5 M EDTA, 5 M NaCl, 1 M DTT, 1 M spermidine and 10% SDS), and the DNA was amplified by PCR and analyzed using Hi-TOM. For single sperm mutation load analysis, individual sperm was sorted by FACS into 96-well plates and lysed using the same oocyte lysis buffer at 37 °C for 30 min, followed by inactivation at 95 °C for 10 min. The DNA was amplified using nested PCR before Hi-TOM analysis. Primers used for nested PCR are listed in the Supplementary Information.

Isolation of immune cells from blood samples

Immune cells from blood samples were isolated by collecting whole blood from the orbital venous plexus using EDTA-K_2-treated tubes. Blood samples were conducted as follows: lysing red blood cells with 1× RBC lysis buffer (eBioscience, 00-4300-54), blocking with Fc block solution (rat anti-mouse CD16/CD32, BD Biosciences, 553142) and staining with a cocktail of fluorescently labeled antibodies for

flow cytometry. The antibody panel included eF450 anti-mouse viability dye, FITC anti-mouse CD11b, PE/Cy7 anti-mouse CD3, BV605 anti-mouse CD19, APC anti-mouse Ly-6C, Alexa Fluor 700 anti-CD45 and BV510 anti-mouse Ly-6G (antibody sources and dilutions specified in the Supplementary Information). Cell populations were identified as follows: B lymphocytes ($CD45^+CD3^-CD19^+$), T lymphocytes ($CD45^+CD3^+CD19^-$), monocytes ($CD45^+CD3^-CD19^-CD11b^+Ly6G^-Ly6C^{hi}$) and neutrophils ($CD45^+CD3^-CD19^-CD11b^+Ly6G^+$). All cell sorting was conducted on a FACSAria Fusion instrument (BD Biosciences). Data were analyzed using FlowJo version 10.4 software.

Assessment of MMP in MEFs and oocytes

MMP in MEFs was determined using TMRM (T668, Invitrogen) following the manufacturer's instructions. In brief, trypsin-digested MEF cells were suspended in PBS and incubated with 100 nM TMRM solution for 30 min at 37 °C. Cells were then gently washed three times with warm PBS. TMRM fluorescence was analyzed by FACS (CytoFLEX LX, Beckman Coulter) at excitation and emission wavelengths of 561 nm and 585 ± 42 nm, respectively, using CytExpert (version 2.4.0.28) and FlowJo software (version 10.4). Data represent three independent experiments, with each experiment analyzing 100,000 cells per group.

For MMP determination in M2 oocytes, oocytes were incubated in 25 nM TMRM solution for 30 min at 37 °C and immediately imaged. Confocal images were captured on a Nikon C2Si microscope using NIS-Elements software (version 5.21).

Mitochondrial ROS production measurement

Mitochondrial ROS production in M2 oocytes was quantified using HKPerox-1 (ref. 62) (MedChemExpress) following the manufacturer's guidelines. In brief, oocytes were co-incubated with HKPerox-1 (10 μM) and CCl_3CN (100 μM) for 30 min at 37 °C, washed three times with HBSS and imaged with the Nikon C2Si microscope (NIS-Elements software, version 5.21).

Immunofluorescent staining of ovarian tissue

Fresh ovarian tissues were fixed overnight in 4% paraformaldehyde (PFA), dehydrated in 30% sucrose buffer, embedded in Optimal Cutting Temperature (OCT) compound and sectioned at 10-μm thickness. Cryosections underwent permeabilization with 0.5% Triton X-100, blocking with 3% BSA for 1 h at room temperature and incubation with a primary antibody followed by appropriate secondary antibodies. High-resolution images and fluorescence quantification were conducted using a Zeiss LSM 800 confocal microscope (Zeiss ZEN software, version 3.4) and ImageJ2 software (version 2.9/1.53t), respectively. Antiobodies used here are listed in the Supplementary Information.

Histological analysis and follicle quantification

Ovarian morphology and follicle quantification were assessed by fixing tissues in Bouin's solution for 5 h, followed by dehydration, embedding and sectioning at 7-μm thickness. Sections were stained with periodic acid–Schiff (PAS) to visualize follicular structures. Follicles were counted on every other section to prevent double counting, focusing on nucleated structures. Follicles were categorized based on developmental stage: primordial (flattened granulosa cells surrounded oocyte), primary (cuboidal granulosa cells surrounded oocyte), secondary (multiple granulosa cell layers surrounded oocyte) and antral (large cavities appeared in follicles).

Transmission electron microscopy analysis

Isolated ovaries were fixed in 2% PFA, 2.5% glutaraldehyde and 0.1 M cacodylate buffer (pH 7.2) at 4 °C overnight, washed with 0.1 M cacodylate buffer and post-fixed with 1% OsO_4 at room temperature for 60 min. After staining with 2% uranyl acetate for 30 min, dehydration was performed with graded ethanol solutions. After replacing the ethanol with acetone and resin, the specimens were embedded in Poly/Bed resin, and 70-nm sections were prepared using a Leica ultramicrotome. These sections were further stained with 2% uranyl acetate and lead citrate to enhance contrast and observed with a C-Twin TEM. Digital images were obtained using a Ceta CCD camera and processed with Talos imaging software (version 3.2.1).

Spatial transcriptomic studies

Preparation of the Stereo-seq chip. The RNA capture chips (D01967F3) were produced by MGI Research and were commercially obtained for this study.

Tissue sectioning, fixation, permeabilization, cDNA synthesis and sequencing. After euthanizing the mice, intact ovarian tissues were collected and embedded in OCT before being rapidly frozen on dry ice and stored at −80 °C. These tissues were then cross-sectioned into 10-μm slices using a CM1950 cryostat (Leica). The sections attached to the Stereo-seq chip were incubated at 37 °C for 3 min, fixed in methanol and stored at −20 °C for 40 min. For comparative histological analysis, the section adjacent to the one on the chip was stained with H&E. The tissue sections on the chip were permeabilized to enable the capture of RNA by DNA nanoball (DNB) probes and reverse transcribed. The cDNA-laden chips were treated with a cDNA release solution and washed with 0.1× SSC buffer. The resulting products were purified and amplified using KAPA HiFi HotStart ReadyMix (Roche). The concentration of cDNA was determined using a Qubit dsDNA Assay Kit (Thermo Fisher Scientific). This cDNA was then used for library preparation and sequencing.

Oocyte identification using spatial transcriptome spots. The initial processing of raw Stereo-seq data was conducted using the SAW pipeline, as previously described[44]. This pipeline facilitated accurate mapping of sequenced reads to their spatial locations in the tissue sections and allowed for precise quantification of gene expression. To address the issue of low RNA capture efficiency at single DNB spots, the raw spatial expression matrix was integrated into larger pseudo-spots using a 50 × 50 window size (bin 50). The resulting GEF file was converted to a Seurat-compatible format using the Python package stereopy[44]. After conversion, the data underwent normalization and were visualized using Seurat's advanced features. For oocyte identification, spots with Gdf9 ≥ 1 and Dppa3 ≥ 1 expression levels were carefully selected for further examination. These spots were then categorized based on their spatial H&E staining coordinates. The mean gene expression values of these spots from individual oocyte were considered as representative of the oocyte gene expression profile.

Differential gene expression analysis. Two distinct methods of differential gene expression analysis were employed. The first method involved ROC analysis, with a threshold set at 0.7. The second method was the Wilcoxon rank-sum test, applying a false discovery rate of 0.28. Both methods are integrated into the Seurat package and were applied with a log fold change threshold of 0.25. The DEGs identified by these methods were then used to independently enrich pathways using GO with the clusterProfiler package[63]. Pathways identified by both analytical approaches were considered significant.

All graphical representations and images in this article, including the intricately detailed heatmaps and plots, were expertly crafted using the R packages Seurat[64], ggplot2 (ref. 65) and ComplexHeatmap[66], ensuring a high level of clarity and visual appeal. The detailed workflow can be referenced in Extended Data Fig. 7.

Rapamycin administration protocol

Rapamycin (T1537, TargetMol) was suspended in PBS containing 0.25% PEG400 and 0.25% Tween 80. *ND1* mutant mice at 10 weeks and 14 weeks of age were administered daily intraperitoneal injections of rapamycin at a dose of 2 mg kg^{-1} body weight per day, for long-term

(49 d) and short-term (21 d) treatment schedules, respectively. An equivalent volume of PBS was injected into 10-week-old mice as experiment control. All mice subjected to treatment were dissected at the age of 17 weeks.

Statistical analysis and reproducibility

Statistical parameters and significance are presented in the figures and figure legends. Statistical analyses were conducted using *t*-test, Mann–Whitney test or Kruskal–Wallis test in GraphPad Prism 9 software (version 9.5). Specific statistical methods for each analysis are specified in the corresponding figure legend. Before analysis, datasets underwent a normality test to ascertain their distribution. For data adhering to a normal distribution, *t*-tests or one-way ANOVA were employed for significance testing. Non-normally distributed data were analyzed using Mann–Whitney or Kruskal–Wallis tests. Data are presented as either the mean ± s.e.m. or the median ± 95% confidence interval (CI).

Sample sizes for animal experiments were not predetermined through statistical methods but were chosen in accordance with sizes reported in similar studies[19,67]. Groups of mice were categorized based on mutation load without employing a randomization strategy for allocation. For Hi-TOM sequencing, reads falling below the threshold of 1,000 were considered low quality and, thus, excluded from analysis. With the exception of this situation, no further data were omitted.

For specific assessments, such as oocyte TMRM quantification (Fig. 3d), oocyte ROS quantification (Fig. 3f) and ovarian follicle counting (Extended Data Fig. 5f), images were anonymized and analyzed by an investigator blinded to group assignments to mitigate bias. In other aspects of the experimental and outcome assessment procedures, blinding of the investigators was not implemented.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All source data are provided with this paper. Stereo-seq data that support the findings of this study have been deposited into STOmis DB of the China National GeneBank DataBase[68] with accession number STT0000087. Further information supporting the findings of this study is available from M.J. upon reasonable request.

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Author contributions

M.J. conceived and led the project. M.J., Y.R., X.D., J.C. and L.Z. designed and conducted most of the experiments. L.Z. and Z.X. constructed the mouse models. Q.L. sorted the immune cells from the blood and collected the sequencing results. J.G., S.L., Z.H. and M.K. assisted with the oocyte assay and Seahorse experiments. M.J. and Y.R. wrote the manuscript, with the help of X.D. and L.Z.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Min Jiang.

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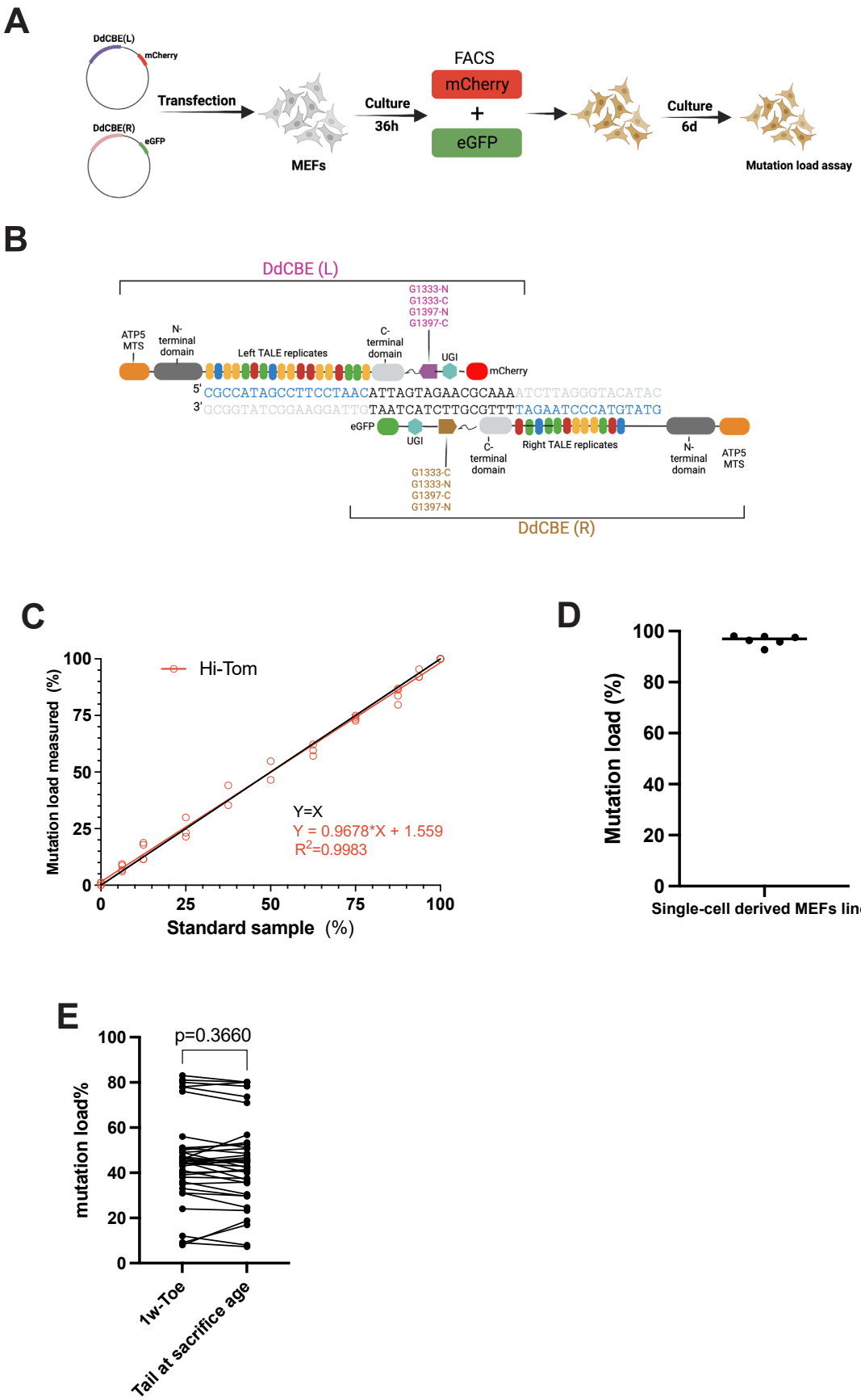
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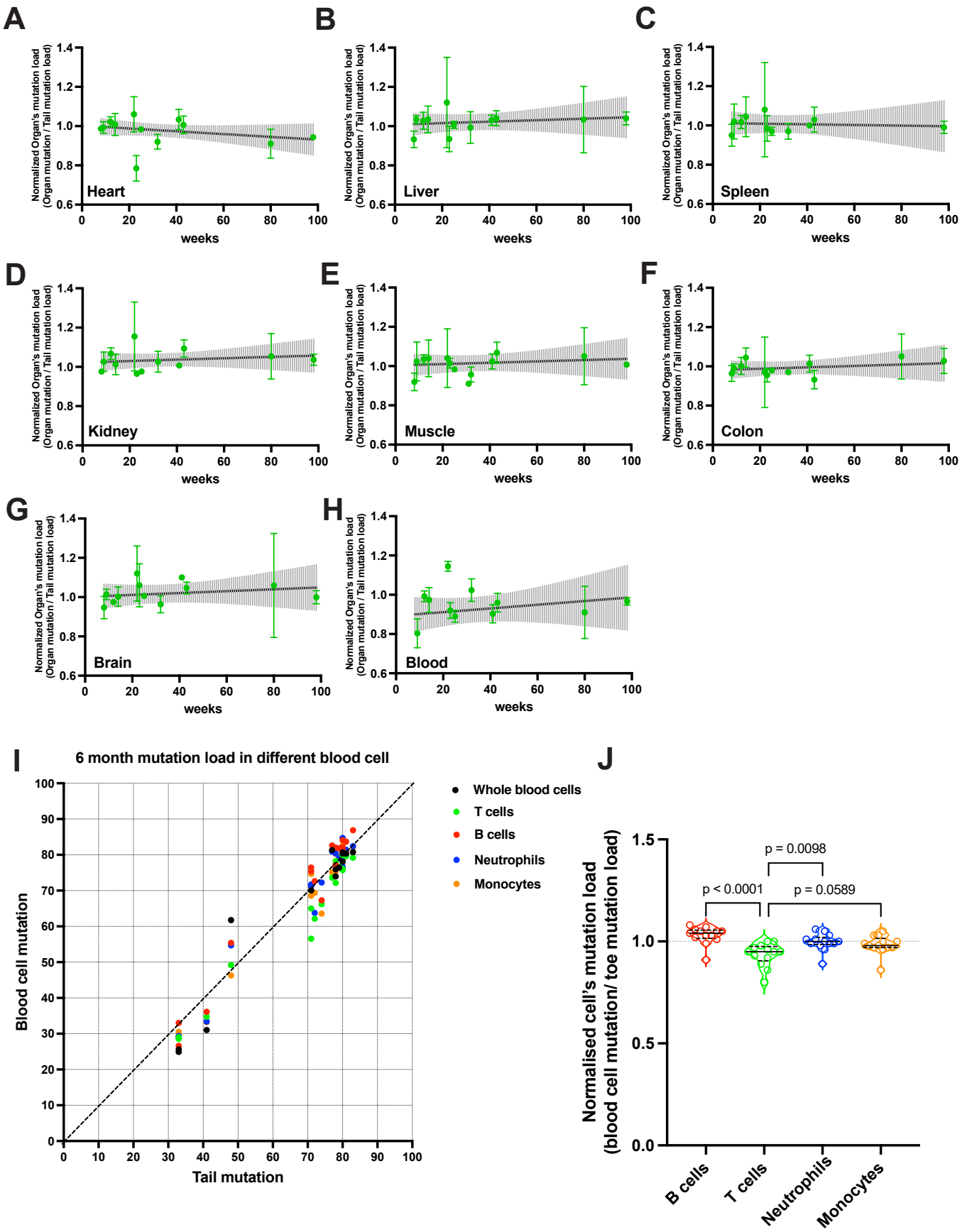


Extended Data Fig. 1 | See next page for caption.

https://doi.org/10.1038/s43587-024-00672-6

Extended Data Fig. 1 | Generation and characterization of mutant MEF lines. (**a**) A schematic outlining the editing process of MEFs and the establishment of mutant, single-cell-derived MEF lines. Created with BioRender.com. (**b**) Depiction of the four pairs of DdCBE editing plasmids targeting the mouse *ND1* gene, highlighting TALE-binding regions in gray-blue and the base editing window in black. Plasmid combinations are labeled according to their left or right positions, with -G1333 or −1397, and with -N or -C termini. Produced using BioRender. (**c**) Validation of Hi-TOM for quantifying mtDNA mutation loads, with the black line representing the line of unity (y = x) and the red dot/line indicating the mutation load of m.2820 G > A as analyzed by Hi-TOM and corresponding linear regression. (**d**) Mutation loads of m.2820 G > A across various MEF cell lines, each originating from individually sorted cell. (**e**) Comparative mutation loads of m.2820 G > A in the toe of 1 week of age and tails at the time of sacrifice (n = 37). Statistical significance assessed using a two-tailed unpaired *t*-test.

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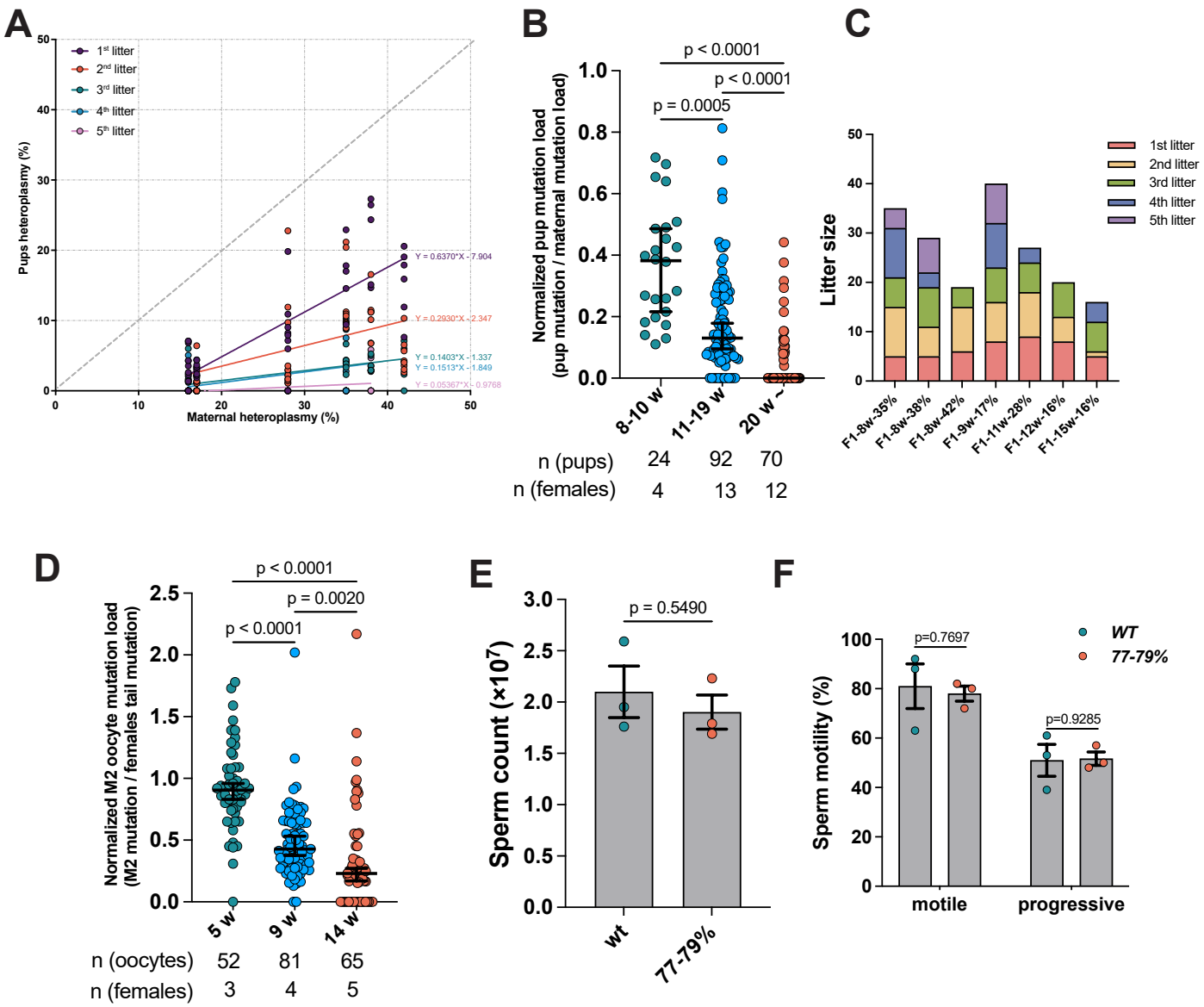


Extended Data Fig. 2 | See next page for caption.

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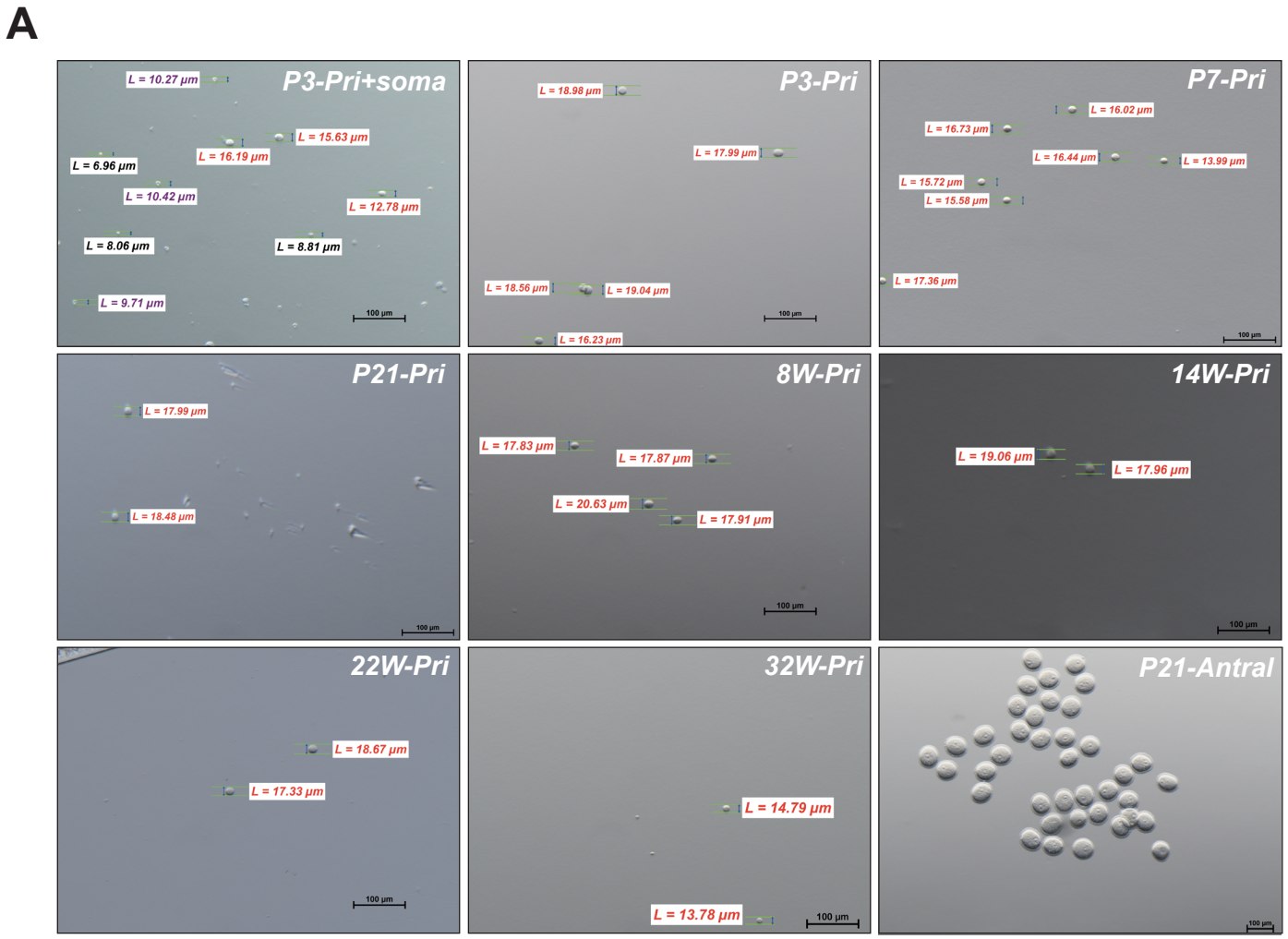
Extended Data Fig. 2 | Tissue-specific *ND1* mutation load variation across multiple ages. (a–h) Mutation loads quantified across various somatic tissues and blood cells in the *ND1* mutant mice: (**a**) Heart; (**b**) Liver; (**c**) Spleen; (**d**) Kidney; (**e**) Muscle; (**f**) Colon; (**g**) Brain; (**h**) Blood. Organ sample sizes are as follows: n = 3 for 8w; n = 3 for 9w; n = 4 for 12w; n = 5 for 14w; n = 2 for 22w; n = 2 for 23w; n = 3 for 25w; n = 4 for 31-32w; n = 3 for 41w; n = 5 for 43w; n = 3 for 80w; n = 3 for 98w. Data are mean ± SEM. **(i–j)** Mutation loads measured in blood and specific sorted cell types, including T cells, B cells, neutrophils, and monocytes. sample sizes: n = 17 animals (I) n = 13 animals (J). *p*-value from one-way ANOVA with Tukey's multiple comparisons test.

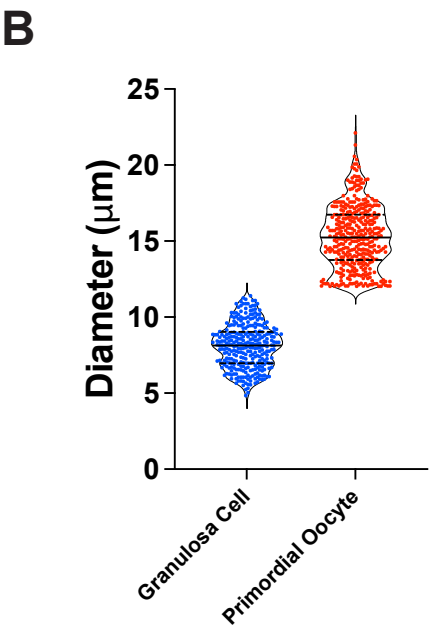
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Extended Data Fig. 3 | Age-enhanced selection on *ND1* mutations also occurs in F1 females. (**a**) Correlation of F2 pup mutation loads (n = 186) with F1 female maternal mutation loads (n = 7), by litter. Different litters are color-coded. The y = x line shows expected transmission without selection, while y=ax+b is the linear regression of the mutation loads of F2 pups from different F1 females but produced in the same litter order. (**b**) Relationship between F1 female maternal age at conception and F2 pup normalized mutation load. Data are median ± 95% CI, with *p*-values from Kruskal-Wallis test and Dunn's comparison. (**c**) Litter sizes of F1 females with varying mutation loads. Different colors indicate distinct litters. (**d**) Mutation load in M2 oocytes from F1 mothers at various ages. Data shown as median ± 95% CI; *p*-value via Kruskal-Wallis test with Dunn's post hoc analysis. **(e-f)** Sperm counts (E) and sperm motility (F) from WT and *ND1* mutant males at 24 weeks of age. Data are mean ± SEM. WT group n = 3; 77–79% mutation load group n = 3. Data are mean ± SEM; with *p*-values from two-tailed unpaired t-test.

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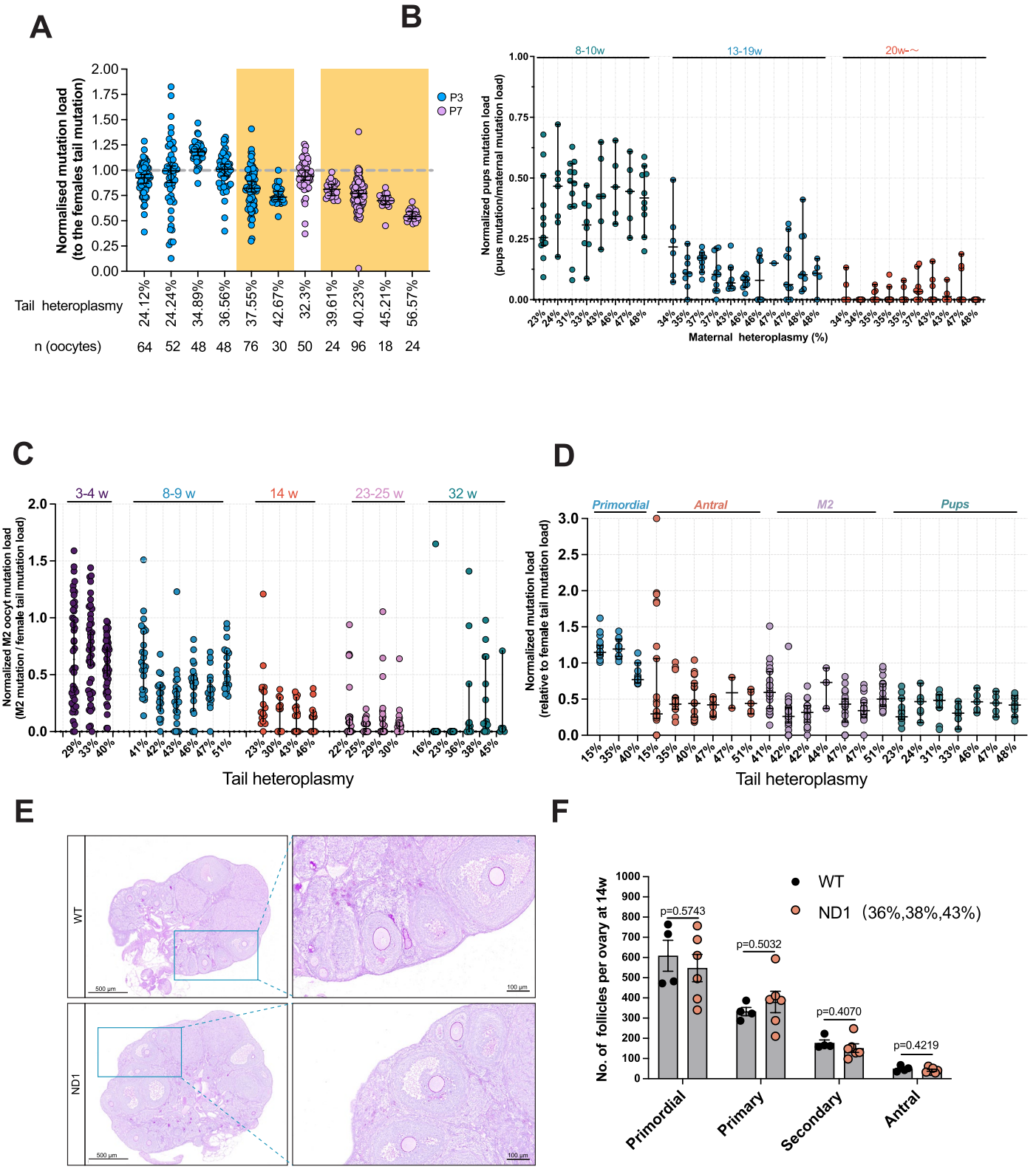


Extended Data Fig. 4 | See next page for caption.

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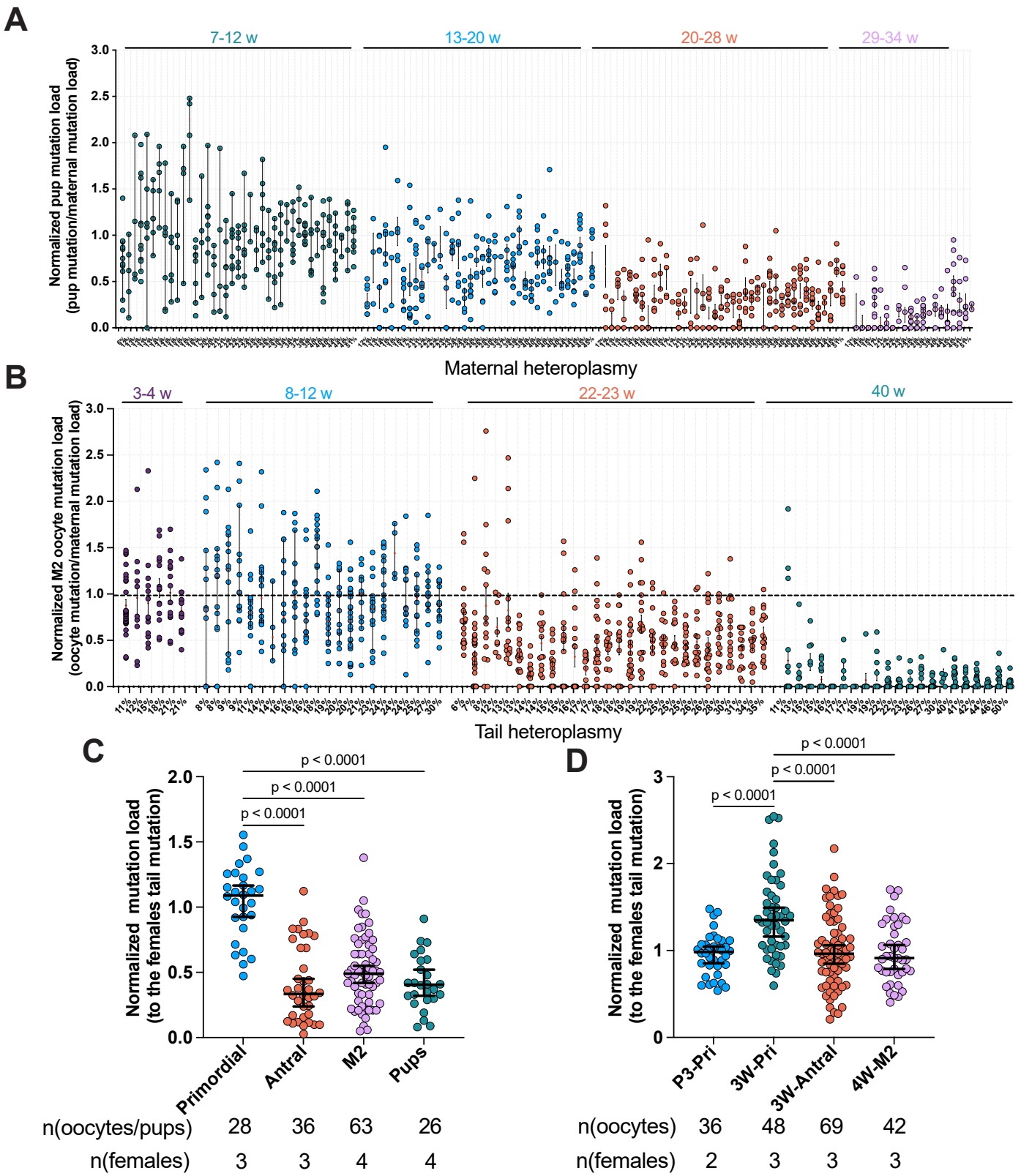
Extended Data Fig. 4 | Morphometry of mouse oocytes in primordial and antral follicles at multiple ages. (**a**) Morphometric analysis of oocytes and surrounding somatic cells in primordial follicles from female mice at various developmental stages (P3, P7, P21, 8 weeks, 14 weeks, 22 weeks, and 32 weeks), and oocytes in antral follicles from P21 females. The diameters of oocytes (red) and somatic cells (soma) were measured. Oocytes are indicated in red, granulosa cells in black and red blood cells in purple. **(b)** The violin plots show the distribution of diameters for oocytes (n = 382 from 2 females) from primordial follicles in P3 mice, and granulosa cells (n = 314 from 2 females), delineating the variability in size of both cell types.

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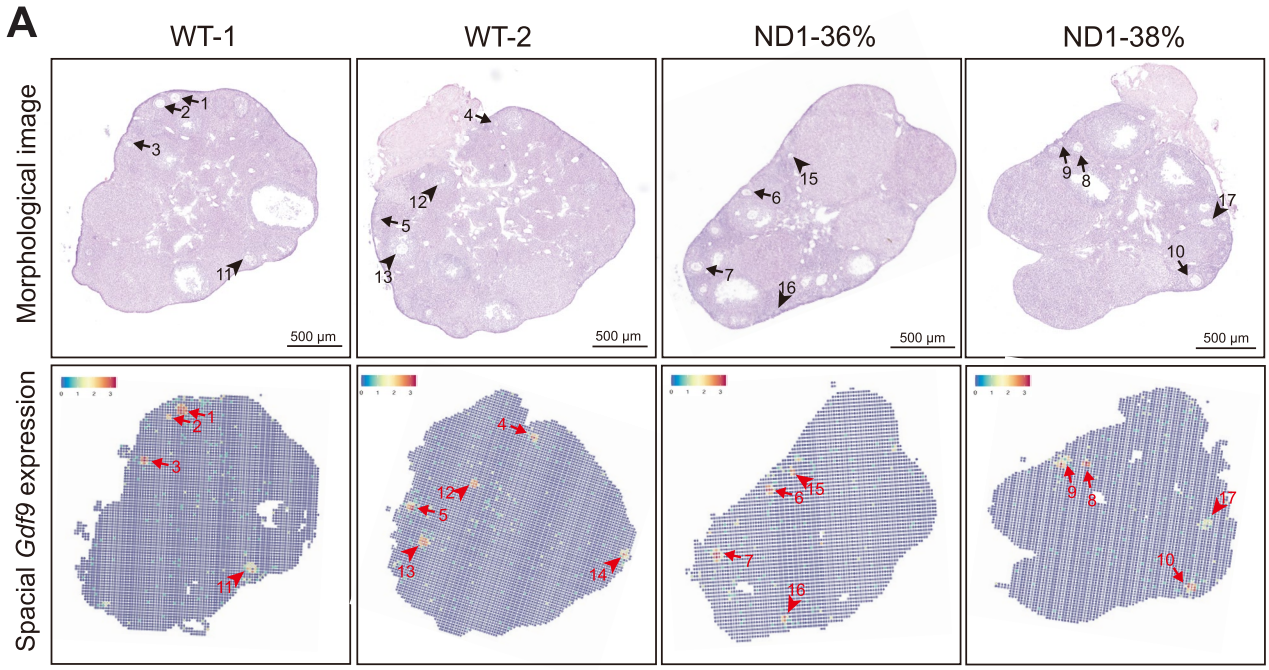
Extended Data Fig. 5 | Age-enhanced purifying selection on m.2820 G > A occurs during oocyte maturation. (**a**) Oocyte mutational loads in primordial follicles of P3 and P7 female mice. Oocytes with mutations >37% are marked in yellow. Individual female mutation loads are shown along the x-axis. Data presented as median ± 95% CI. P3 values correspond to those in Fig. 4e, now with expanded individual data. (**b**) Mutation loads in pups born to females of varying ages. Data are median ± 95% CI.This dataset correspond to those in Fig. 2e, with in-depth individual details. Female mutation loads are displayed along the x-axis. (**c**) Mutation load in M2 oocytes from females of different ages. Data are median ± 95% CI. This dataset correspond to those in Fig. 3a, with in-depth individual details. Female mutation loads are displayed along the x-axis. (**d**) Mutation loads throughout oocyte development and in pups from individual mothers. Data are median ± 95% CI. This dataset correspond to those in Fig. 4b, with in-depth individual details. Female mutation loads are displayed along the x-axis. (**e**) Representative PAS-stained ovarian section images from 14-week-old *ND1* and WT mouse. (**f**) Counted follicle numbers at various developmental stages in ovarian sections from 14-week-old *ND1* (n = 6 ovaries) and WT (n = 4 ovaries) mice. Data are mean ± SEM; with *p*-values from two-tailed unpaired t-test.

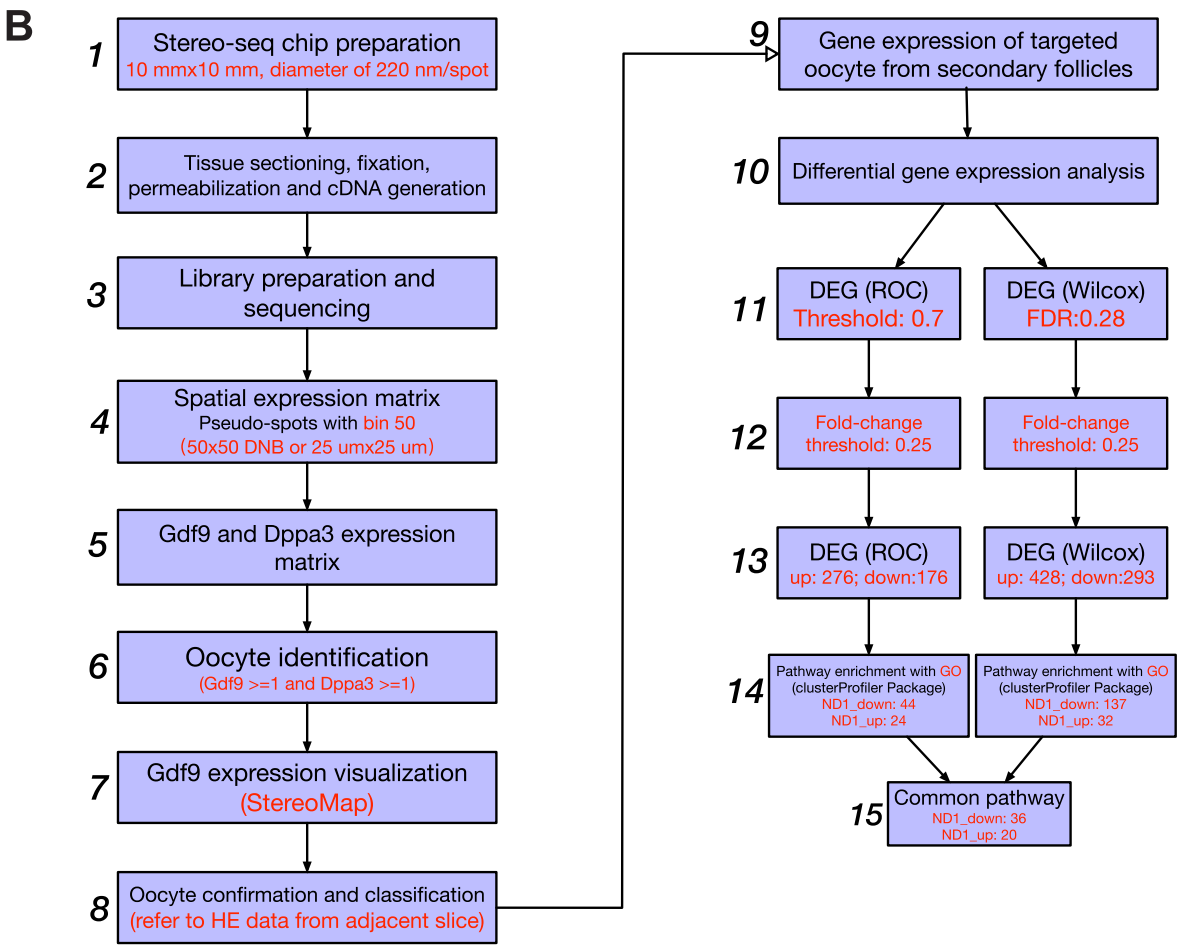
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Extended Data Fig. 6 | Age enhances purifying selection on the mouse m.12918 G > A mutation during maternal transmission. (**a**) Mutation loads in pups born to individual females of varying ages. Data are median ± 95% CI. This dataset correspond to those in Fig. 6e, with in-depth individual details. Female mutation loads are displayed along the x-axis. (**b**) Mutation load in M2 oocytes from individual females of varying ages. Data are median ± 95% CI. This dataset correspond to those in Fig. 6g, with in-depth individual details. Female mutation loads are displayed along the x-axis. (**c**) Normalized mutation loads throughout oocyte development and in pups from multiple females at 22-week-old. Data are median ± 95% CI; with *p*-value from Kruskal-Wallis test followed by Dunn's multiple comparisons. (**d**) Normalized mutation loads at different oocyte development and M2 from young females under 4 weeks old. Data are median ± 95% CI; with *p*-value from one-way ANOVA followed by Tukey's multiple comparisons.

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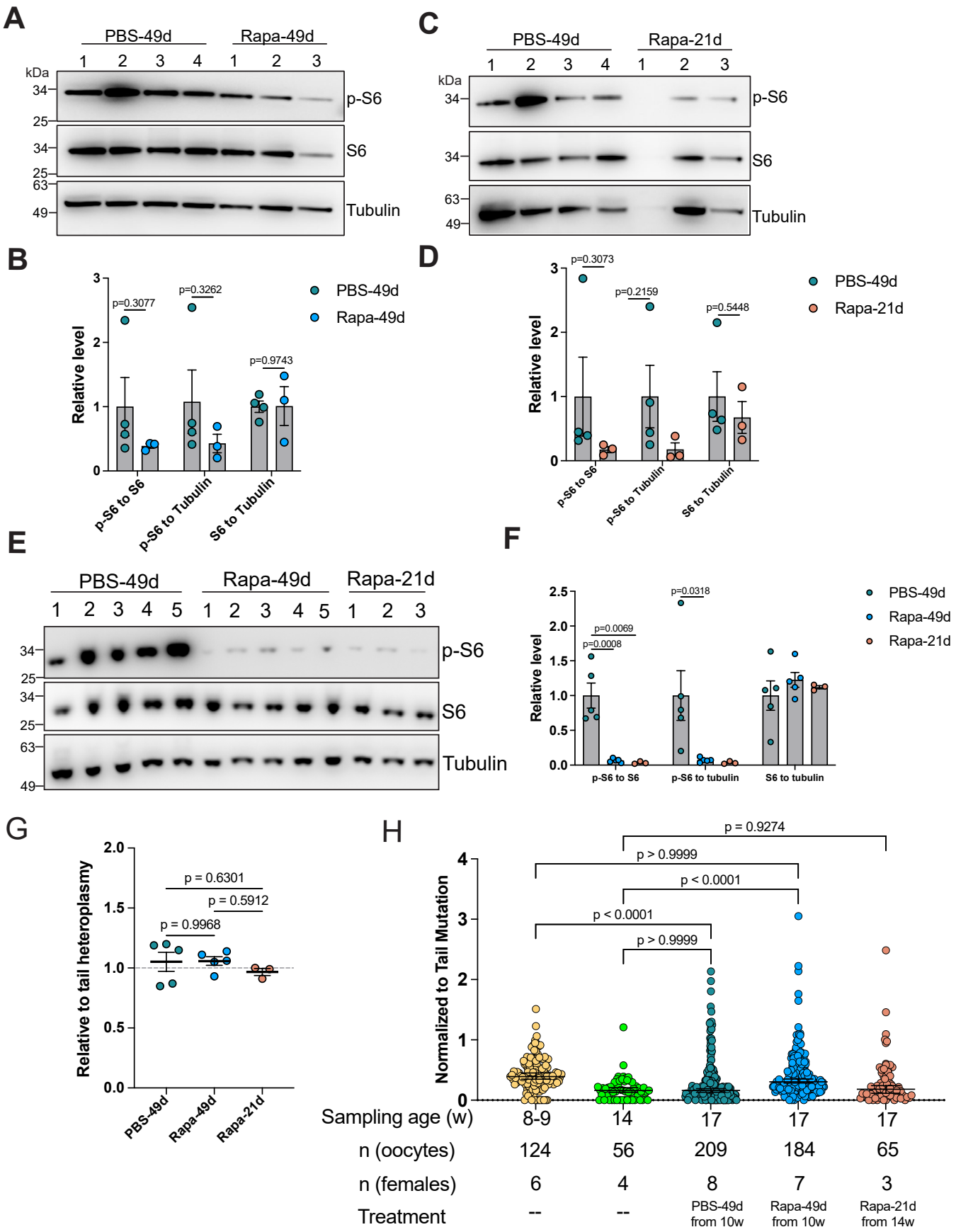


Extended Data Fig. 7 | See next page for caption.

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Extended Data Fig. 7 | Spatial transcriptomic analysis of the ovary. (**a**) Top row: H&E staining of ovarian sections from *ND1* mutant and WT mice. Bottom row: Gdf9 gene expression profile obtained from spatial transcriptomics on adjacent sections for Stero-seq. Arrows indicate oocytes in secondary follicles (#1-10); while arrowheads point to oocytes in other follicles (#11, #12, #13, #16, #17), follicle-free regions (#15) or no mached follicle region (#14) in top H&E image (for detail in Supplementary Table 3). (**b**) Schematic diagram of the spatial transcriptomics analysis workflow applied in this study, detailing the step-by-step process from initial tissue sectioning to the final analysis and data interpretation.

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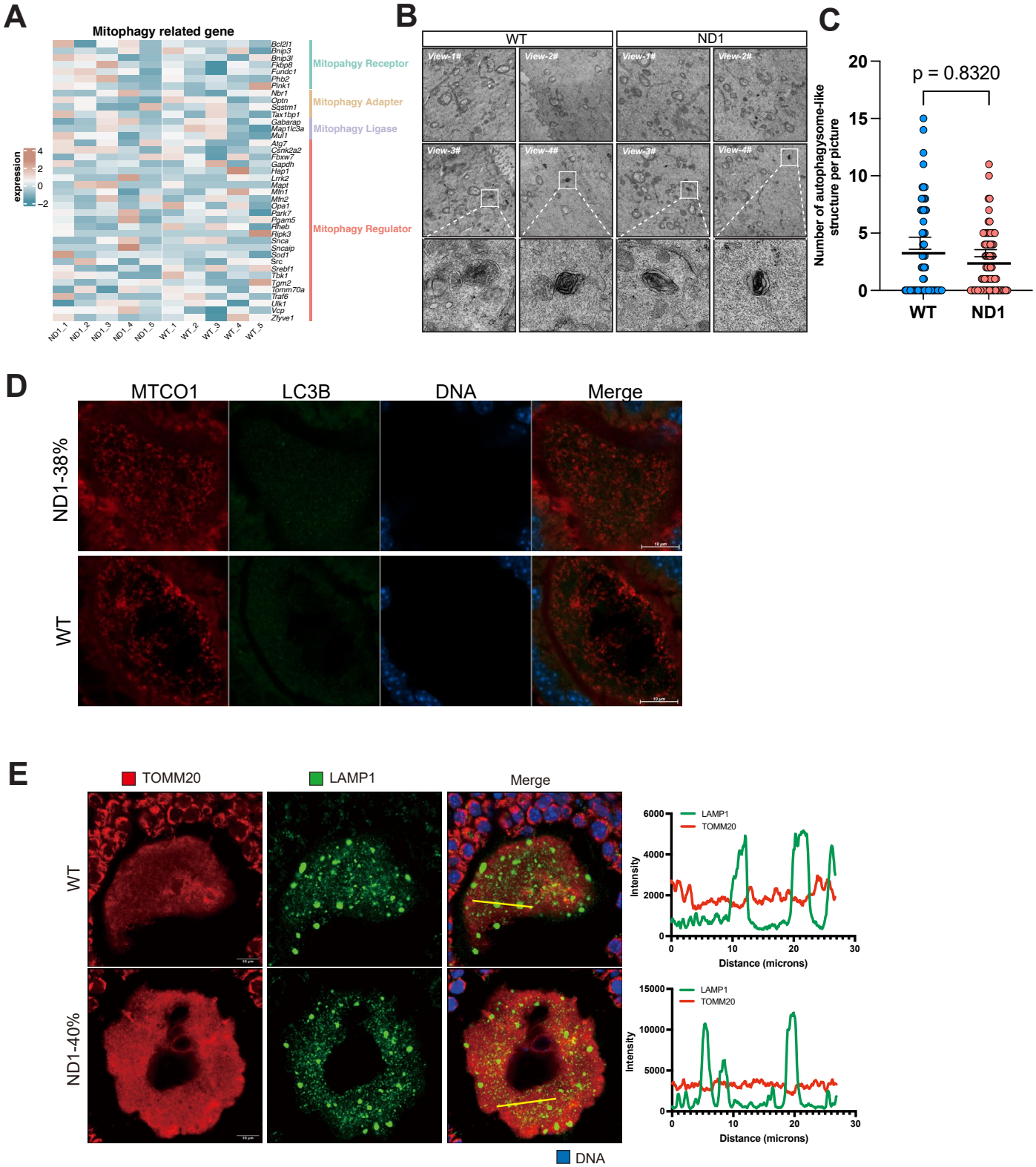


Extended Data Fig. 8 | See next page for caption.

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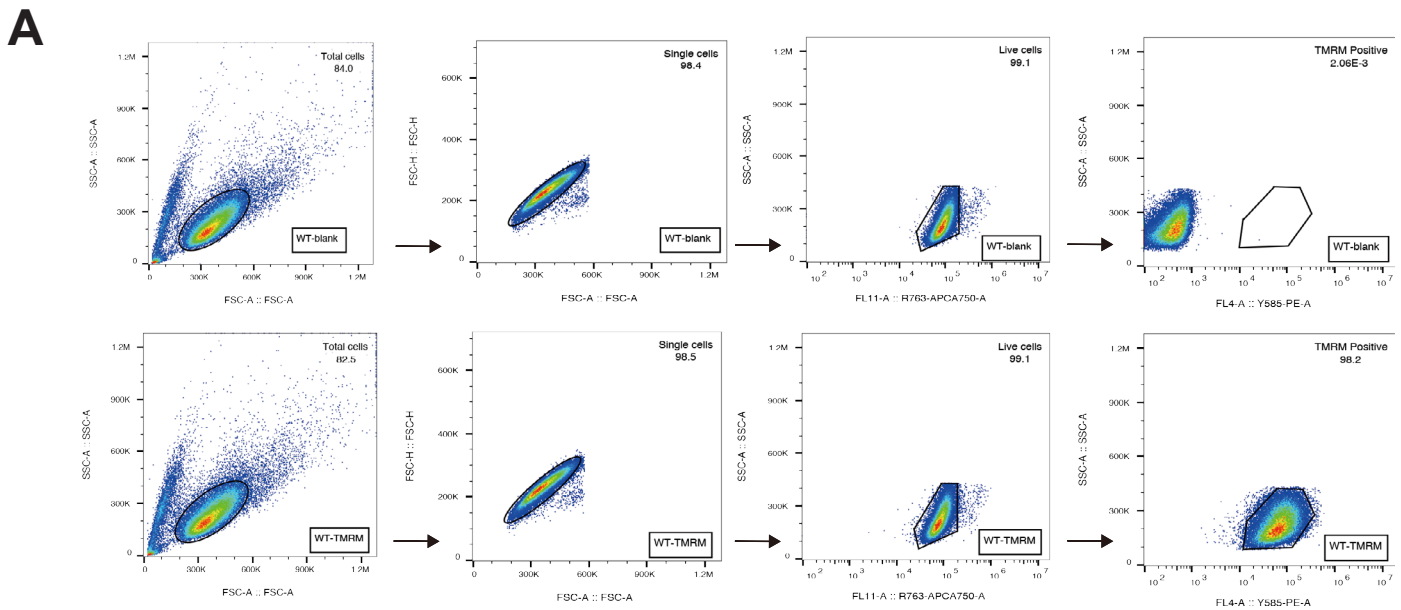
Extended Data Fig. 8 | Detailed assessment of rapamycin treatment efficacy in ovary and heart. (**a**) Representative western blot of p-S6 levels in ovarian tissue following long-term treatment of either PBS or rapamycin in mice. (**b**) Quantitative analysis of (A). n = 4 for PBS-49d; n = 3 for Rapa-49d. Data are presented as the mean ± SEM, with *p*-values from two-tailed unpaired *t*-test. (**c**) Representative western blot of p-S6 levels in ovarian tissue following short-term treatment of either PBS or Rapamycin in mice. (**d**) Quantitative analysis of (C). n = 4 for PBS-49d; n = 3 for Rapa-21d. Data are mean ± SEM, with *p*-values from two-tailed unpaired *t*-test. (**e**) Representative western blot of p-S6 levels in heart tissue following short-, long-term treatment of rapamycin and PBS in mice. (**f**) Quantitative analysis of (E). n = 5 for PBS-49d; n = 5 for Rapa-49d; n = 3 for Rapa-21d. Data are mean ± SEM, with *p*-value from two-tailed unpaired *t* test. (**g**) The mutation load in heart tissues from mice treated with either PBS or Rapamycin. n = 5 for PBS-49d; n = 5 for Rapa-49d; n = 3 for Rapa-21d. Data are mean ± SEM, with *p*-values from two-tailed unpaired *t*-test. (**h**) Comparision of normalized mutational loads in M2 oocytes from treated (the same dataset as those in Fig. 8i) or non-treated mice (the same dataset as those in Fig. 3a). Each symbol represents an individual oocyte. Data are median ± 95% CI, with *p*-values via Kruskal-Wallis test and Dunn's post hoc analysis.

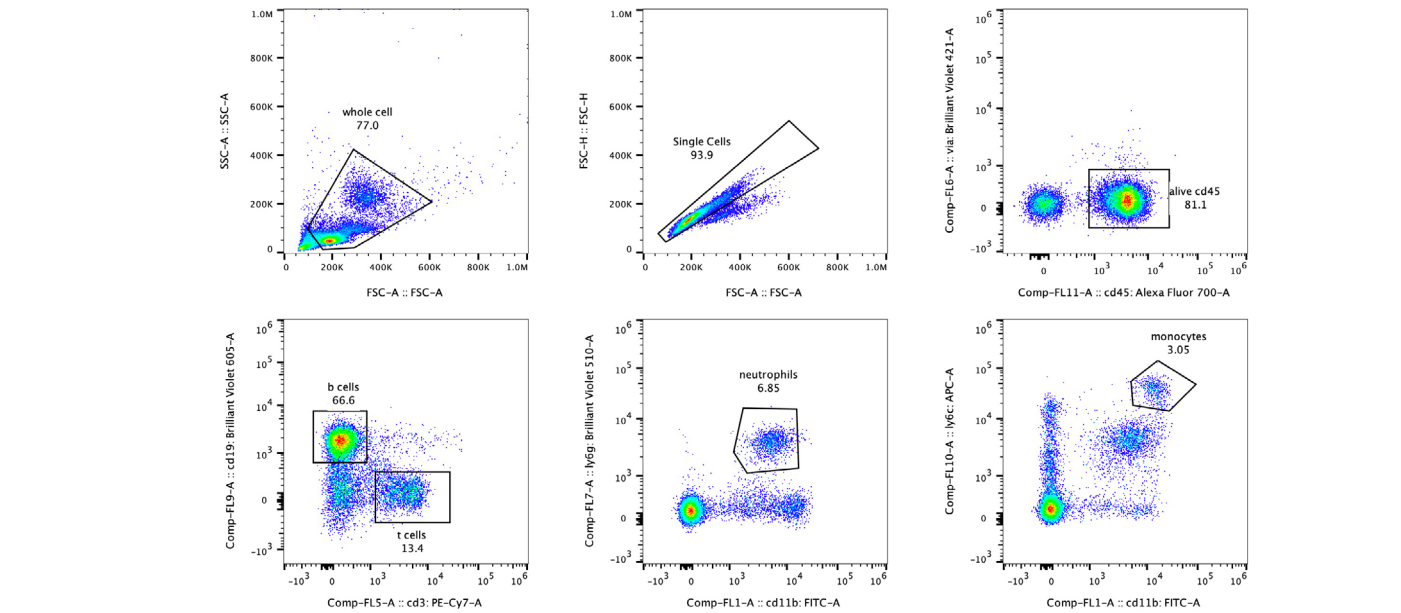
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Extended Data Fig. 9 | No observable mitophagy in the *ND1* mutant oocytes. (**a**) Heatmap presenting detailed gene expression profiles related to mitophagy pathway from Stero-seq analysis, log-normalized for clarity using Seurat. (**b**) Representative TEM image of oocytes in secondary follicles from *ND1* and WT mice. Four distinct views from each mouse are presented. A zoomed-in view highlights a potential autophagic structure. (**c**) Quantitative analysis of autophagysome-like structures in (B). Data are mean ± SEM. n = 60 for WT; n = 71 for *ND1*; with *p*-value from two-tailed Mann Whitney test. (**d**) Representative images of LC3B and MTCO1 co-staining in oocytes in secondary follicle of WT and *ND1* mutant mice. The experiment was repeated three times. (**e**) Representative images of LAMP1 and TOMM20 co-staining in oocytes in secondary follicle of WT and *ND1* mutant mice. The experiment was repeated five times.

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Extended Data Fig. 10 | Flow cytometry gating strategies. (**a**) TMRM staining Gating: Initial gating excluded debris via FSC-A vs SSC-A, followed by doublet exclusion using FSC-A vs FSC-H. Finally, eFluor 780 positive dead cells were removed from the analysis. (**b**) Immune cell isolation Gating: Debris was first excluded using FSC-A vs SSC-A, with doublets subsequently removed via FSC-A vs FSC-H. Specific immune cell populations were identified using the following markers: B lymphocytes ($CD45^+CD3^-CD19^+$), T lymphocytes ($CD45^+CD3^+CD19^-$), Monocytes ($CD45^+CD3^-CD19^-CD11b^+Ly6G^-Ly6C^{hi}$), and Neutrophils ($CD45^+CD3^-CD19^-CD11b^+Ly6G^+$).

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Software and code

Policy information about availability of computer code

Data collection	CytExpert software (V2.4.0.28) WAVE (V2.6) Nikon elements software (V5.21) ZEISS ZEN software (v3.4) Velox (V3.2.1)
Data analysis	IVOS sperm motility analyser (V12.3A) WAVE (V2.6) FlowJo (V10.4) ImageJ2(V2.9/1.53t) Graphpad Prism (V9.5.1) R version (V4.3.0)

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All source data are provided in the source data file. Stereo-seq data that support the findings of this study have been deposited into STOmis DB of China National GeneBank DataBase (CNGBdb) with accession number STT0000087.

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Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

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All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes for animal experiments were not predetermined through statistical methods but were chosen in accordance with sizes reported in comparable studies (1. Freyer, C. et al. Nat Genet 44, 1282–1285 (2012). 2. Zhang, H. et al. Sci Adv 7, eabi5657.). Sample sizes used for each experiments is provided in the figures and figure legends.
Data exclusions	For Hi-TOM sequencing, reads falling below the threshold of 1000 were considered low quality and thus excluded from analysis. With the exception of this situation, no further data was omitted.
Replication	All attempts at replication were successful and were used in the analysis to generate statistical values. The in vitro experiments were highly reproducible. Each experiments was repeated at 3 independent occasions unless otherwise stated. The in vivo experiment was performed on separate occasions, which were all successful and included in the analysis in order to ensure the reproducibility.
Randomization	Samples were categorized according to mutation load, and randomization was not employed.
Blinding	For specific assessments, such as oocyte TMRM quantification (Figure 3D), oocyte ROS quantification (Figure 3F), and ovarian follicle counting (Extended Data Figure 5F), images were anonymized and analyzed by an investigator blind to group assignments to mitigate bias. In other aspects of the experimental and outcome assessment procedures, blinding of the investigators was not implemented.

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☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

1.Total OXPHOS Rodent WB Antibody Cocktail, mouse mAbs,Abcam, Cat# ab110413
(anti-NDUFB8 antibody: clone 20E9DH10C12; anti-SDHB antibody: clone 21A11AE7; anti-UQCRC2 antibody: clone 13G12AF12BB11; anti-MTCO1 antibody: clone 1D6E1A8; anti-ATP5A antibody: clone 15H4C4)
2.anti-VDAC1, mouse monoclonal antibody, clone N152B/23, Merck, Cat# MABN504
3.anti-beta-Actin, mouse monoclonal antibody, clone AC-15, Sigma-Aldrich, Cat# A1978
4.anti-beta-Tubulin, rabbit monoclonal antibody, Beyotime, Cat# AF1216 (Suppliers cannot provide clone numbers)
5.anti-phospho-S6 (ser235/236),rabbit monoclonal antibody, clone D57.2.2E, CST, Cat#4858
6.anti-total-S6, rabbit monoclonal antibody,clone 5G10, CST, Cat#2217
7.anti-TOMM20, rabbit polyclonal antibody, Abcam, Cat#ab78547
8.anti-MTCO1, mouse monoclonal antibody, clone 1D6E1A8, Abcam, Cat#ab14705
9.anti-LAMP1, rat monoclonal antibody, clone 1D4B, SantaCruz, Cat#sc-19992
10. FITC anti-mouse/human CD11b, rat monoclonal antibody, clone M1/70, Biolegend, Cat#101205
11. PE/Cyanine7 anti-mouse CD3, rat monoclonal antibody, clone 17A2, Biolegend, Cat#100220
12. BV605 anti-mouse CD19, rat monoclonal antibody, clone 6D5, Biolegend, Cat#115539
13. APC anti-mouse Ly-6C, rat monoclonal antibody, clone HK1.4, Biolegend, Cat#128016
14. Alexa Fluor 700 anti-mouse CD45, rat monoclonal antibody, clone 30-F11, Biolegend Cat#103128
15. BV510 anti-mouse Ly-6G, rat monoclonal antibody, clone 1A8, Biolegend Cat#127633
16. anti-LC3B, mouse monoclonal antibody, clone E5Q2K, CST, Cat#83506

Validation

Antibodies were chosen from literature and validated through western blot for specificity. All antibodies are commercially available. Additional information on validation can be found on the manufacturer's websites, see below.
1. Total OXPHOS Rodent WB Antibody Cocktail, Abcam, Cat# ab110413, validated by PMID: 37913525, reactivity: mouse; applications: WB.
2. anti-VDAC1, mouse monoclonal antibody, Merck, Cat# MABN504, validated by PMID: 25688136, reactivity: mouse; applications: WB.
3. anti-beta-Actin, mouse monoclonal antibody,Sigma-Aldrich, Cat# A1978, validated by PMID: 31276497, reactivity: mouse; applications: WB.
4. anti-beta-Tubulin, rabbit monoclonal antibody, Beyotime, Cat# AF1216, validated by PMID: 31614141, reactivity: mouse; applications: WB.
5. anti-phospho-S6 (ser235/236),rabbit monoclonal antibody,CST, Cat#4858, validated by PMID: 38177284 and PMID: 37516825, reactivity: mouse, applications: IF and WB.
6.anti-total-S6, rabbit monoclonal antibody,CST, Cat#2217, validated by PMID: 38035266, reactivity: mouse; applications: WB.
7.anti-TOMM20, rabbit polyclonal antibody, Abcam, Cat#ab78547, validated by PMID: 35417232, reactivity: mouse; applications: WB.
8.anti-MTCO1, mouse monoclonal antibody, Abcam, Cat#ab14705, validated by PMID: 36253618, reactivity: mouse; applications: WB.
9.anti-LAMP1, rat monoclonal antibody, SantaCruz, Cat#sc-19992, validated by PMID: 36280666, reactivity: mouse; applications: IF.
10. FITC anti-mouse/human CD11b, rat monoclonal antibody, Biolegend, Cat#101205, validated by PMID: 35305295, reactivity: mouse; applications: flow cytometry.
11.PE/Cyanine7 anti-mouse CD3, rat monoclonal antibody, Biolegend, Cat#100220, validated by PMID: 29887373, reactivity: mouse; applications: flow cytometry.
12.BV605 anti-mouse CD19, rat monoclonal antibody, Biolegend, Cat#115539, validated by PMID: 21436069, reactivity: mouse; applications: flow cytometry.
13. APC anti-mouse Ly-6C, rat monoclonal antibody, Biolegend, Cat#128016, validated by PMID: 29861385, reactivity: mouse; applications: flow cytometry.
14. Alexa Fluor 700 anti-CD45, rat monoclonal antibody, Biolegend Cat#103128, validated by PMID: 29628141, reactivity: mouse; applications: flow cytometry.
15. BV510 anti-mouse Ly-6G, rat monoclonal antibody, Biolegend Cat#127633, validated by PMID: 32555459, reactivity: mouse; applications: flow cytometry.
16. anti-LC3B, mouse monoclonal antibody, CST, Cat#83506, validations were provided on the manufacturer's website, "knockout validated".

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)

MEF cell line was purchased from ATCC (CRL-2907).

Authentication	STR profiling was performed to authenticate the MEF cell line.
Mycoplasma contamination	Cell lines were regularly tested using PCR Mycoplasma detection kit (TransDetect, Cat# FM311-01). None of the cell lines used in this study tested positive for Mycoplasma.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this study.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	1. For establishment of ND1 and ND5 mutant mouse , 3-week-old C57B6/J female mice were superovulated and mated with 8-12-week-old male C57BL/6J mice to collect zygote. The microinjected zygotes were transfered into the oviducts of 8-10-week-old ICR pseudopregnant female mice at 0.5 days postcoitum. 2.For producing pups from young ND1 mutant females (aged 3–4 weeks), the superovulation protocol was repeated as above, and post-mating with wild-type C57BL/6J males, the zygotes were collected. These were allocated into two groups: the Assisted Reproductive Technology (ART) group, wherein zygotes were directly transferred to the oviducts of pseudopregnant ICR mice, and the second-round-editing group, where zygotes were subjected to another round of DdCBE mRNA microinjection (at a concentration of 250 ng/μl) before transfer to the ICR recipients. 4. For oocytes collection, P3 to 40w-old mutant females were used. 5. For sperm and testis sample collection, 25-43 week old mutant males were used. The mice were housed under a 12-hour light-dark cycle at controlled ambient temperature (22 ± 1 °C) and relative humidity (40%-70%). Adequate standard chow and water were available ad libitum via cage lids. Animal ages and sexes are provided in the manuscript and figure legends.
Wild animals	Study did not involve wild animals.
Reporting on sex	Maternal findings in this study are applicable only to females.
Field-collected samples	Study did not involve samples collected from the field.
Ethics oversight	All procedures involving animals were conducted in strict accordance with the guidelines for the Care and Use of Laboratory Animals of Westlake University and received approval from the Institutional Animal Care and Use Committee at Westlake University (Approval No. AP#22-003-JM-3).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	1. For membrane potential measurement of MEFs, cells were digested with trypsin and suspended in PBS. Then, the cells were incubated in 100 nM TMRM solution for 30 min at 37°C and gently washed three times with warm PBS containing

	Fixable Viability Dye eFluor 780 (1:1000 dilution) . The cells were resuspended in PBS containing and applied to FACS. 2. For isolation of immune cells from blood samples, whole blood cells were obtained from the orbital venous plexus using EDTA-K2 treated tubes. To generate single-cell suspensions, red blood cells were lysed with 1X RBC lysis buffer (eBioscience™, 00-4300-54) on ice for 5 minutes. The pellets were then resuspended in 50 µl of Fc block solution (purified rat anti-mouse CD16/CD32, diluted 1:200, BD Cat#553142) and incubated on ice for 5 minutes. For staining, the cells were incubated on ice for 30 minutes in the dark. Antibody solution contains the following fluorescently labeled antibodies: eF450 anti-mouse viability dye (diluted 1:1000, Thermofisher Cat#65-0863-14), FITC anti-mouse CD11b (diluted 1:200, Biolegend Cat#101205), PE/Cy7 anti-mouse CD3 (diluted 1:200, Biolegend Cat#100220), BV605 anti-mouse CD19 (diluted 1:200, Biolegend Cat#115539), APC anti-mouse Ly-6C (diluted 1:200, Biolegend Cat#128016), Alexa Fluor 700 anti-CD45 (diluted 1:200, Biolegend Cat#103128), and BV510 anti-mouse Ly-6G (diluted 1:200, Biolegend Cat#127633). After staining, the pellets were washed once and resuspended for sorting using flow cytometry.
Instrument	Flow cytomytry was performed on a Beckman Coulter CytoFLEX LX.
Software	CytExpert v2.4 and FlowJo v10 were used for data acquisition and analysis.
Cell population abundance	Cell populations were confirmed by running sorted populations on the Beckman Coulter CytoFLEX LX. For immune cells sorting, the population of each cell type: B-cell 66.6%, T-cell 13.4%, Neutrophils: 6.85%, Monocytes: 3.05%.
Gating strategy	The gating strategy in flow cytometry experiments consisted of debris exclusion by FSC-A vs SSC-A analysis and subsequent doublet exclusion by FSC-A vs FSC-H analysis. Dye eFluor 780 positive dead cells were then excluded. The following panels were used to classify blood cells: B lymphocytes: CD45+CD3-CD19+, T lymphocytes: CD45+CD3+CD19-, Monocytes: CD45 +CD3-CD19-CD11b+Ly6G-Ly6Chi, Neutrophils: CD45+CD3-CD19-CD11b+Ly6G+. The gating strategy are included in Extended Data Fig.10.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.